



Last Time

- Chinese Word Processing
- Automatic Chinese Word Segmentation
- Unknown Word Identification
- Unknown Word Normalization



Today's class

- Word Meanings
 - Computer Expression of Word Meanings
 - English Semantic Resources
 - Chinese Semantic Resources
 - Applications of Semantic Resources
- Word Sense Disambiguation
 - Background and Restrictions
 - Automatic Word Sense Disambiguation
 - Knowledge-based Approach
 - Machine Learning based Approach
 - Hybrid Approach
 - Evaluations



Word Meaning/Sense

- In linguistics, a word sense is one of the meanings of a word
- In each sentence, we associate a different meaning of the observing word based on hints the rest of the sentence
 - We went to see the **play** *Romeo and Juliet* at the theater
 - The children went out to **play** *in the park*
- Semantics: Study of word meanings

Expression and organization of Word Sense



- A word meaning is expressed by a description string in a dictionary
- Word Entries in different dictionary
 - 翻腾
 - Sense 1: 上下滚动: 波浪~
 - Sense 2: 翻动: 几个柜子都~到了
 - 现代汉语词典
 - Description string
 - Examples
 - Glosses: Definitions and/or example sentences

Computer Expression of Word Meaning



- Described by **Sememe 义元**:
 - the smallest unit of meaning and is not further divisible
 - sememe <move> is shared by the various action morphemes walk, run, roll, jump, etc
- Described by definition
 - 翻腾
 - DEF=exercise | 锻炼,sport | 体育
 - DEF=roll | 滚
 - DEF=turn | 扭转
- Described by Synonyms Set with glosses
 - WordNet
 - **good, right, ripe**
 - (most suitable or right for a particular purpose; "a good time to plant tomatoes"; "the right time to act"; "the time is ripe for great sociological changes")
 - CiLin
 - **翻腾 叨登 捣腾 倒腾 翻倒 倾倒 扇风**



Terminology

- **Homonyms 同音**

- one of a group of words that share the same spelling and the same pronunciation but have different meanings
- *Bank* a financial institution / river bank.

- **Polyseme 多义:**

- words or phrases with multiple, related meanings
- *Bed* in “river bed” / a household furniture.
- Bank: financial institute, building of the financial
- institute, storage of blood (blood bank)

- **Synonyms 同义**

- different words having similar or identical meanings
- *student / pupil; buy / purchase*

- **Synset 同义词集**

- a set of one or more synonyms



Terminology

- **Antonyms 反义**

- different words having contradictory or contrary meanings
- *Perfect/imperfect big/small*

- **Hypernymy 上位**

- the semantic relation of being super-ordinate or belonging to a higher rank or class
- 'living things' is a hypernym of 'human'

- **Hyponymy 下位**

- the semantic relation of being sub-ordinate or belonging to a lower rank or class
- 'adult' is a hyponym of 'human'.

- **Coordinate terms 同类**

- Y is a coordinate term of X if X and Y share a hypernym
- *wolf* is a coordinate term of *dog*



Terminology

- **Holonymy 整体**

- a word that defines the relationship between a term denoting the whole and a term denoting **a part of, or a member of**, the whole
- 'tree' is a holonym **of** 'bark树皮', 'trunk树干'

- **Meronymy 部份**

- a word that names **a part of** a larger whole
- 'finger' is a meronym of 'hand'
- 'wheel' is a meronym of 'automobile'.

- **Metonymy 转指**

- a figure of speech in which a concept is referred to by the name of something closely associated with that concept
- *White House vs. President Beijing vs. China*



Terminology

- **Troponym**方式

- the verb Y is a troponym of the verb X if the activity Y is doing X in some manner
- to **lisp** 发音 is a troponym of to **talk**

- **Entailment**蕴涵

- the verb Y is entailed by X if by doing X you must be doing Y
- to **sleep** is entailed by to **snore**



WordNet

- WordNet is a lexical database for the English language
- Groups English words into synsets, provides
- short, general definitions, and records the various semantic relations between these synonym sets
- Purposes:
 - Produce a combination of dictionary and thesaurus that is more intuitively usable
 - Support automatic text analysis and artificial intelligence applications



WordNet Information

- Development began in 1985 at the Cognitive Science Laboratory of Princeton University
- WordNet's latest version is 3.0.
- The database contains 155,287 words organized in 117,659 synsets for a total of 206,941 word-sense pairs
- In compressed form, it is about 12 megabytes in size



WordNet: Design

- Organize the lexical knowledge according to word senses rather word form
- Description Target in WordNet
 - compound 复合词
 - phrasal verb 短语动词
 - collocation 搭配词
 - idiomatic phrase 成语
 - word 单词
- Word is the minimum unit
 - No morpheme
 - No frames

WordNet: Lexical Knowledge and World Knowledge



- Lexical knowledge:
 - Maintained in dictionary
 - 'hit' : a hard blow delivered with the hand or something held in it
 - 'hit' is verb
 - 'hit' is followed by the object
- World Knowledge
 - Maintained in Encyclopedia
 - a violent impact between things
 - Object should be solid / no gas
- WordNet focus on Lexical Knowledge
 - In some entries, combine lexical knowledge and world knowledge



WorNet Organization

- WordNet distinguishes
 - nouns, verbs, adjectives and adverbs
 - follow different grammatical rules
- Every synset contains a group of synonymous words or collocations
- Different senses of a word are in different synsets
- The meaning of the synsets is further clarified with short defining glosses
- Synsets are connected to other synsets via a number of semantic relations



WorNet: Sysnet Example

- **Synset**: Good, right, ripe
- **Glosses**: most suitable or right for a particular purpose
- **Example collocations**:
 - "a good time to plant tomatoes"
 - "the right time to act"
 - "the time is ripe for great sociological changes"
- Different meanings are in the different synsets



WorNet: Polysemes

POS	Monosemy and Senses	Polysemes	Senses of Polysemes
Noun	94685	14510	40002
Verb	5920	5168	18221
Adjectives	15981	5479	15175
Adverb	3820	787	1900
Total	120406	25944	75298

POS	Senses (including monosemy)	Senses
Noun	1.23	2.75
Verb	2.17	3.52
Adjectives	1.45	2.76
Adverb	1.24	2.41



Format of WordNet Entries

The noun “bass” has 8 senses in WordNet.

1. bass¹ - (the lowest part of the musical range)
2. bass², bass part¹ - (the lowest part in polyphonic music)
3. bass³, basso¹ - (an adult male singer with the lowest voice)
4. sea bass¹, bass⁴ - (the lean flesh of a saltwater fish of the family Serranidae)
5. freshwater bass¹, bass⁵ - (any of various North American freshwater fish with lean flesh (especially of the genus Micropterus))
6. bass⁶, bass voice¹, basso² - (the lowest adult male singing voice)
7. bass⁷ - (the member with the lowest range of a family of musical instruments)
8. bass⁸ - (nontechnical name for any of numerous edible marine and freshwater spiny-finned fishes)

The adjective “bass” has 1 sense in WordNet.

1. bass¹, deep⁶ - (having or denoting a low vocal or instrumental range)
 ”a deep voice”; ”a bass voice is lower than a baritone voice”;
 ”a bass clarinet”



WorNet: Semantic Relations - Noun

- **Hypernyms 上位**: Y is a hypernym of X if every X is a (**kind of**) Y
 - Canine is a hypernym of dog
- **Hyponyms 下位**: Y is a hyponym of X if every Y is a (**kind of**) X
 - dog is a hyponym of canine
- **Coordinate terms 同类**: Y is a coordinate term of X if X and Y share a hypernym
 - wolf is a coordinate term of dog
- **Holonym 整体**: Y is a holonym of X if X is a **part of** Y
 - building is a holonym of window
- **Meronym 部分**: Y is a meronym of X if Y is a **part of** X
 - window is a meronym of building



WorNet: Knowledge Structure

- Both nouns and verbs are organized into hierarchies, defined by hypernym

dog, domestic dog, Canis familiaris

=> *canine, canid*

=> *carnivore*

=> *placental, placental mammal, eutherian, eutherian mammal*

=> *mammal*

=> *vertebrate, craniate*

=> *chordate*

=> *animal, animate being, beast, brute, creature, fauna*

- At the top level, these hierarchies are organized into base types
 - 25 primitive groups for nouns, and 15 for verbs



Primitive groups for nouns

{act, action, activity}

{animal, fauna}

{artifact}

{attribute, property}

{body, corpus}

{cognition, knowledge}

{communication}

{event, happening}

{feeling, emotion}

{food}

{group, collection}

{location, place}

{motive}

{natural object}

{natural phenomenon}

{person, human being}

{plant, flora}

{possession}

{process}

{quantity, amount}

{relation}

{shape}

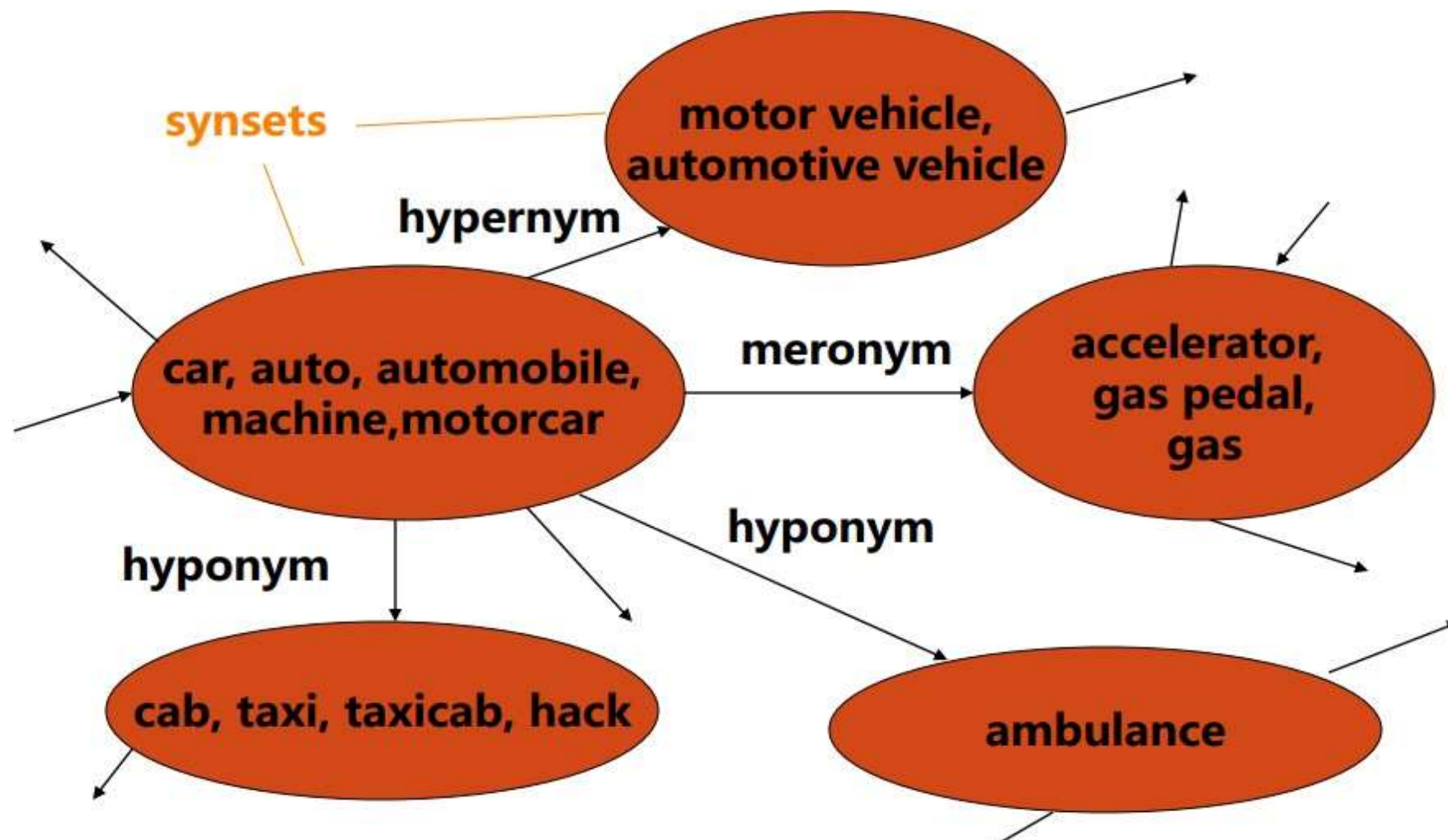
{state, condition}

{substance}

{time}



A WordNet Snapshot



WorNet: Semantic Relations - Verb



- **Hypernym**: the verb Y is a hypernym of the verb X if the activity X is a (**kind of**) Y
 - to perceive 感觉 is an hypernym of to listen
- **Troponym**方式: the verb Y is a troponym of the verb X if the **activity Y is doing X in some manner**
 - to lisp 发音 is a troponym of to talk
- **Entailment**蕴涵: the verb Y is entailed by X if by **doing X you must be doing Y**
 - to sleep is entailed by to snore
- **Coordinate terms**: those verbs sharing a common hypernym
 - to lisp and to yell (叫喊)



Primitive groups for Verbs

- 1) 身体动作动词 (Verbs of Bodily Care and Functions)
- 2) 变化动词 (Verbs of Change)
- 3) 通信动词 (Verbs of Communication)
- 4) 竞争动词 (Competition Verbs)
- 5) 消费动词 (Consumption Verbs)
- 6) 接触动词 (Contact Verbs)
- 7) 认知心理动词 (Cognition Verbs)
- 8) 创造动词 (Creation Verbs)
- 9) 运动动词 (Motion Verbs)
- 10) 情感心理动词 (Emotion or Psych Verbs)
- 11) 状态动词 (Stative Verbs)
- 12) 感知动词 (Perception Verbs)
- 13) 领属动词 (Verbs of Possession)
- 14) 社会交互 (Verbs of Social Interaction)
- 15) 气象动词 (Weather Verbs)

WorNet: Semantic Relations - Adjectives



- Described as N-D space
- Synset of adjective: Adjective Cluster
- Adjective clusters are connected through **Antonyms** relation
- Similar to relation 近似: ponderous/heavy
- Direct Antonyms: light/heavy
- Indirect Antonyms: light/ ponderous

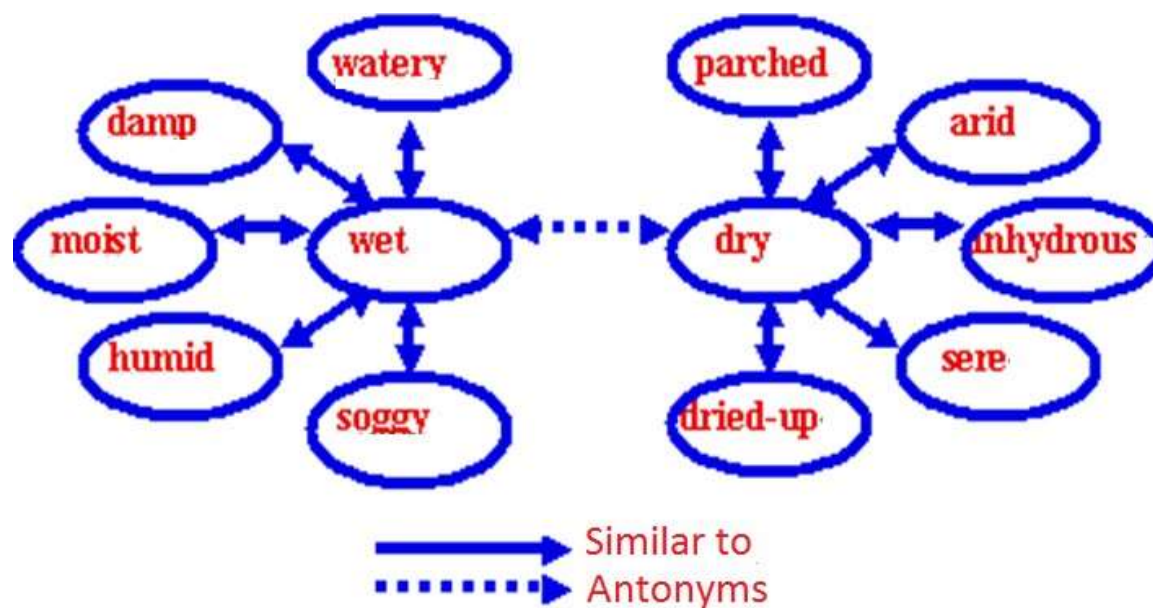
Sense 2

ponderous -- (having great mass and weight and unwieldiness; 'burden'; "ponderous weapons")

=> heavy (vs. light) -- (of comparatively great physical weight; "heavy metal"; "heavy mahogany furniture")



WordNet: Adjective Cluster





WorNet as Thesaurs

- WordNet is organized based on Synset
- Synset is not the only expression for concept
- Synet is connected by semantic relations
 - Hypernyms
 - Hyponyms
 - Coordinate terms
 - Holonym
 - Meronym 部分
 - ...



WorNet as Ontology

- Lexical Ontology
 - The hypernym/hyponym relationships among the noun synsets
 - Must be corrected before being used since hundreds of basic semantic inconsistencies
 - transforming WordNet into a lexical ontology
 - distinguishing the specialization relations into `subtypeOf` and `instanceOf` relations
 - associating intuitive unique identifiers to each category
- Part of the integration of WordNet 1.7 into the knowledge base of WebKB-2



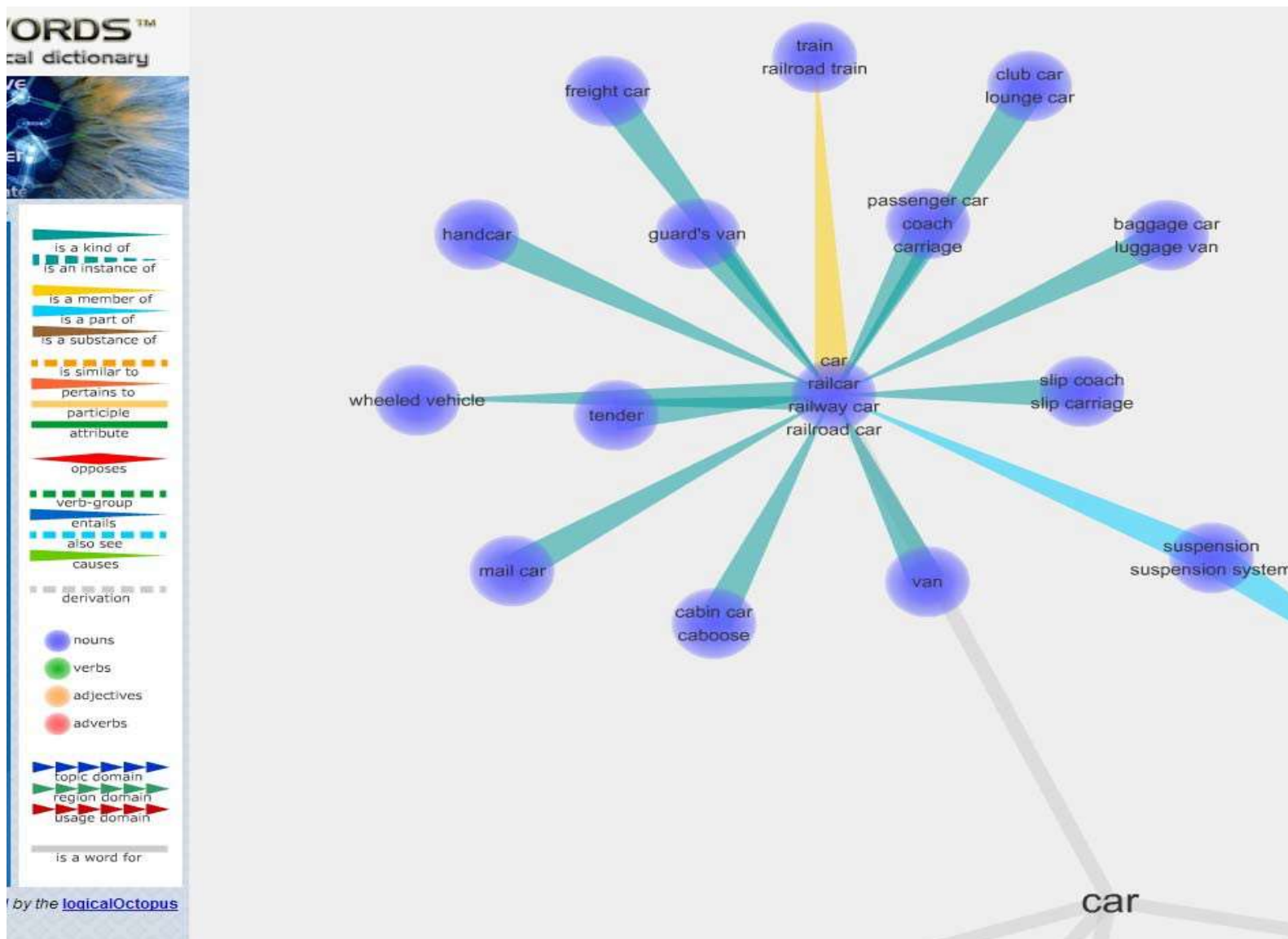
WordNet 3.0 Statistics

POS	Unique Strings	Synsets	Total Word-Sense Pairs
Noun	117,798	82,115	146,312
Verb	11,529	13,767	25,047
Adjective	21,479	18,156	30,002
Adverb	4,481	3,621	5,580
Totals	155,287	117,659	206,941



WordNet 3.0 Visualization

Visualization: www.visuwords.com



Word Similarity Estimation based on WordNet



- Basic Idea: Two words(synset) which are more adjacent or have common hierarchy in WordNet, are more similar.

$$\text{sim}_{\text{path}}(c_1, c_2) = -\log \text{pathlen}(c_1, c_2)$$

$$\text{sim}_{\text{Resnik}}(c_1, c_2) = -\log P(\text{LCS}(c_1, c_2))$$

$$\text{sim}_{\text{Lin}}(c_1, c_2) = \frac{2 \times \log P(\text{LCS}(c_1, c_2))}{\log P(c_1) + \log P(c_2)}$$

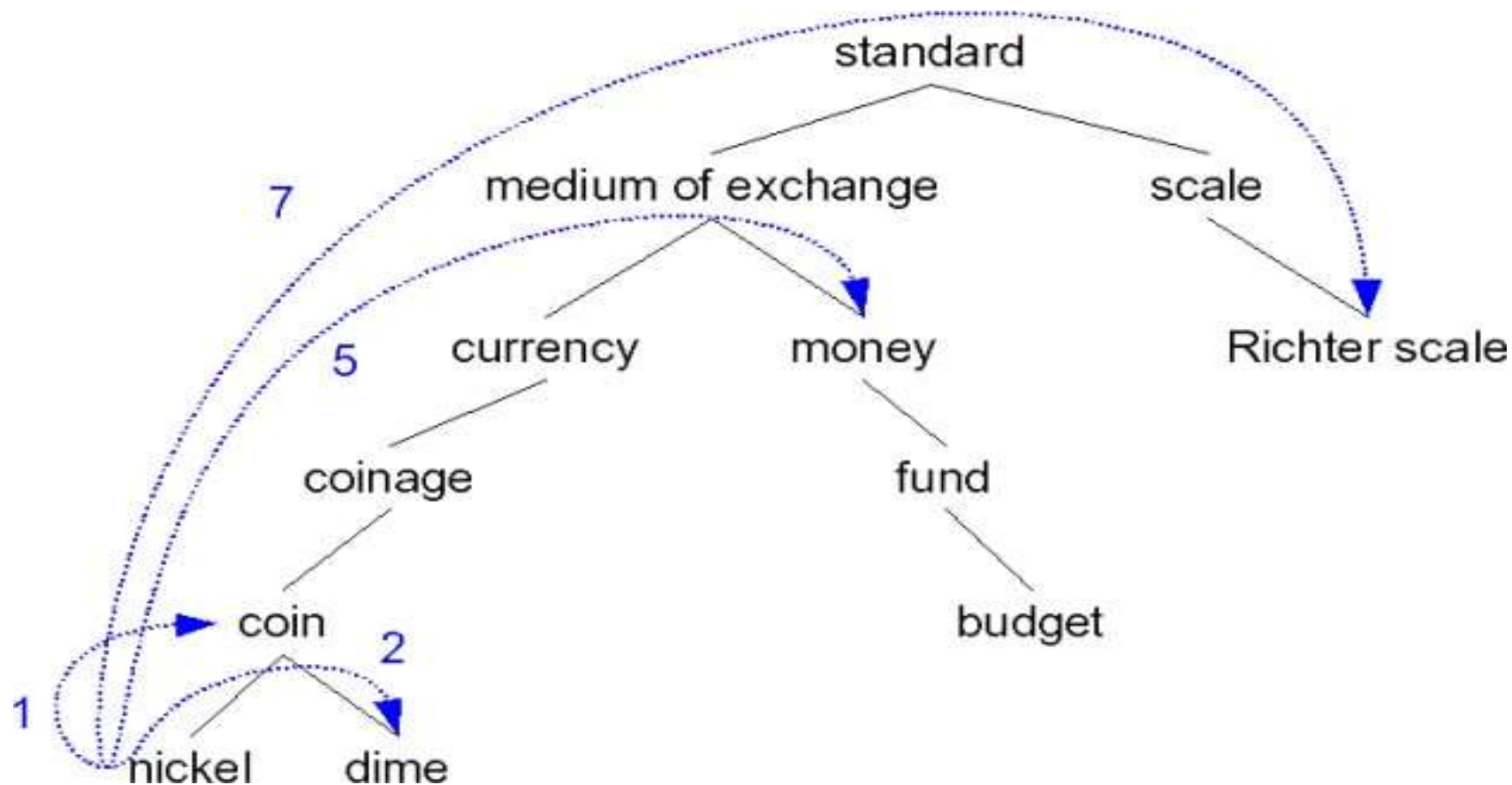
$$\text{sim}_{\text{jc}}(c_1, c_2) = \frac{1}{2 \times \log P(\text{LCS}(c_1, c_2)) - (\log P(c_1) + \log P(c_2))}$$

$$\text{sim}_{\text{eLesk}}(c_1, c_2) = \sum_{r, q \in \text{RELS}} \text{overlap}(\text{gloss}(r(c_1)), \text{gloss}(q(c_2)))$$



Example 1: Path based similarity

- The shorter path in WordNet hierarchy, the greater similarity.



Example 1: Improvements of Path based similarity



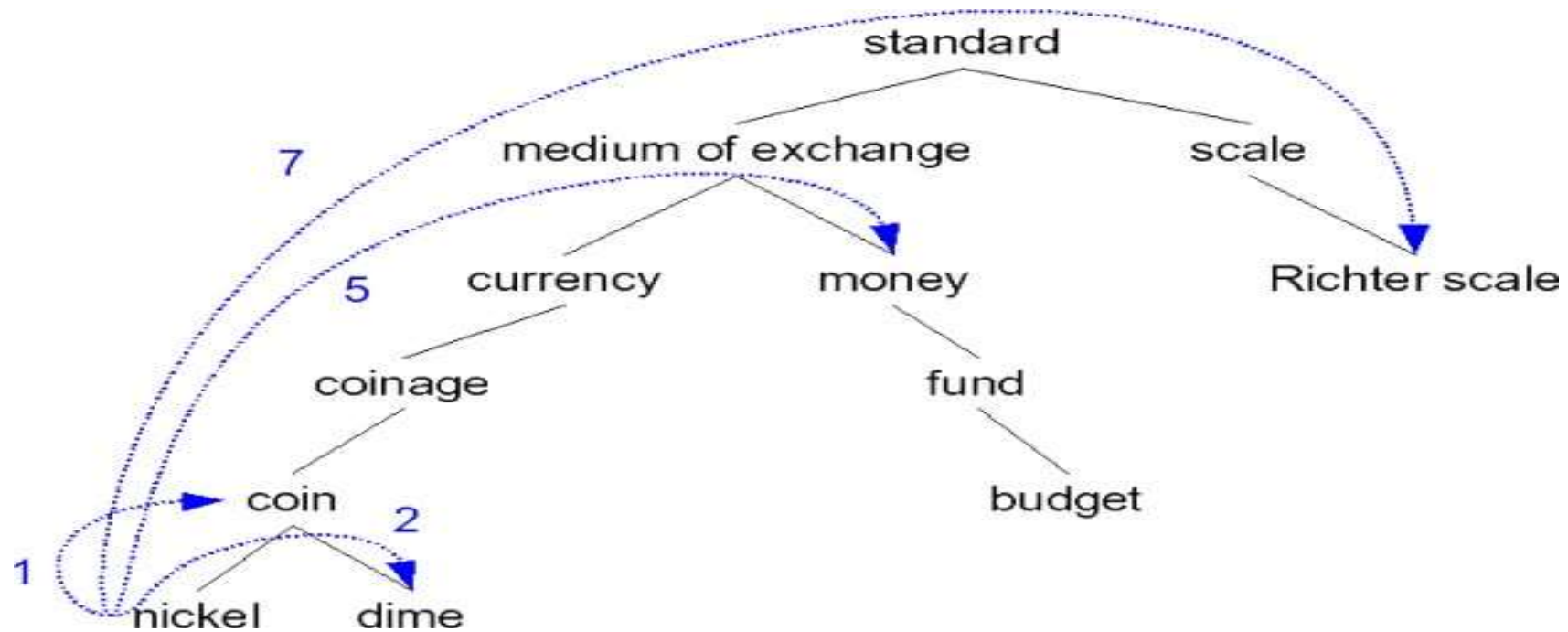
- $\text{pathlen}(c1, c2)$ = the number of edge in the the shortest path between $c1$ and $c2$
- $\text{simpath}(c1, c2) = -\log \text{pathlen}(c1, c2)$
- $\text{wordsim}(w1, w2) =$

$$\max_{c1 \in \text{senses}(w1), c2 \in \text{senses}(w2)} \text{sim}(c1, c2)$$

Example 1: Problem of Path based similarity



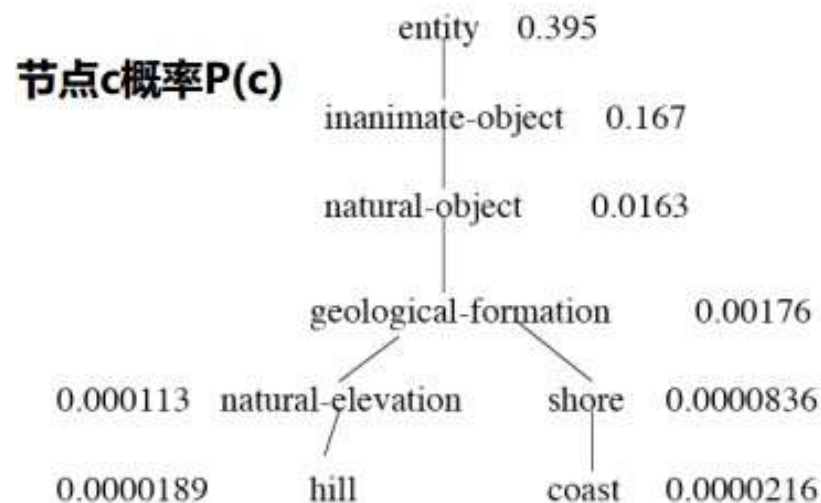
- Assuming each link(edge) represents the same distance
- Nickel to money and nickel to standard have the same similarity?





Example 2: Resnik method

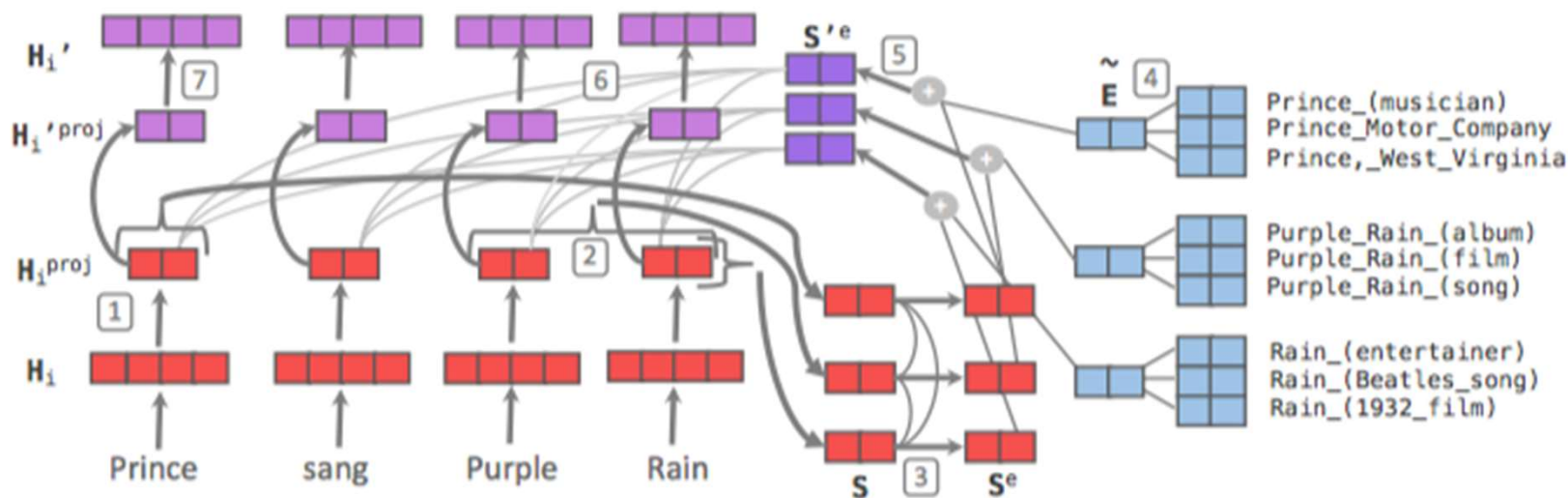
- Use the information content of the lowest common ancestor node to measure the similarity.
- $\text{sim}_{\text{resnik}}(c1, c2) = -\log P(\text{LCS}(c1, c2))$
- The lower common ancestor node, the higher similarity.





Applications of WordNet

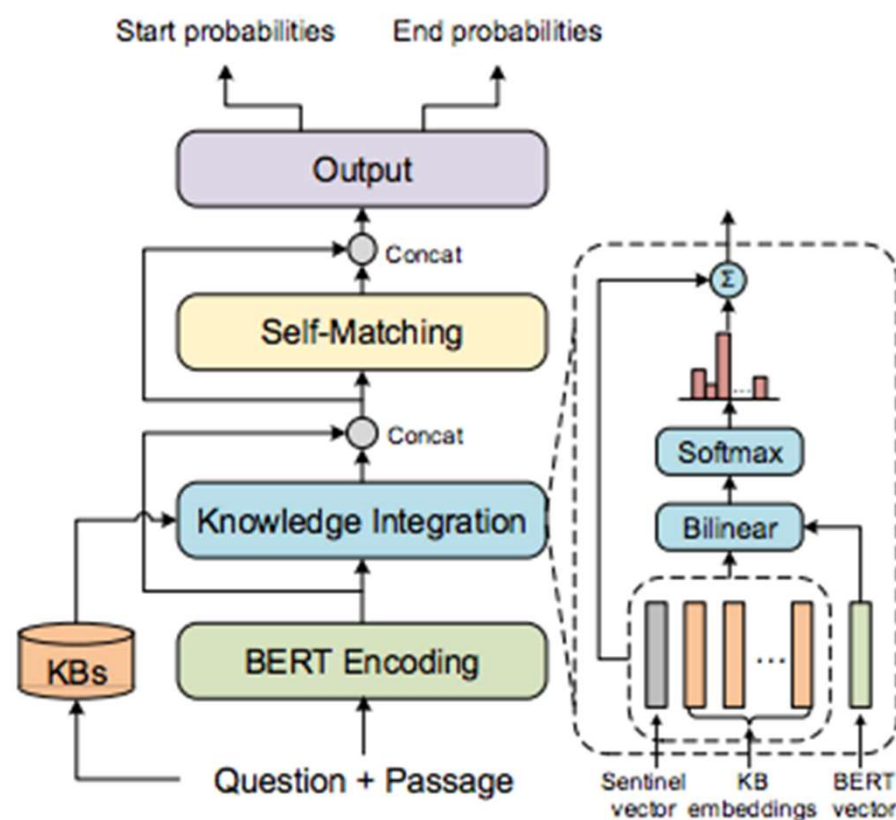
- Knowledge Enhanced Contextual Word Representations
 - [Peters et al. 2019] propose a general method to embed multiple knowledge bases (KBs) into large scale models, and thereby enhance their representations with structured, human-curated knowledge.





Applications of WordNet

- Machine reading comprehension
 - [Yang et al. 2019] investigate the potential of leveraging external knowledge bases (KBs) to further improve BERT for MRC.



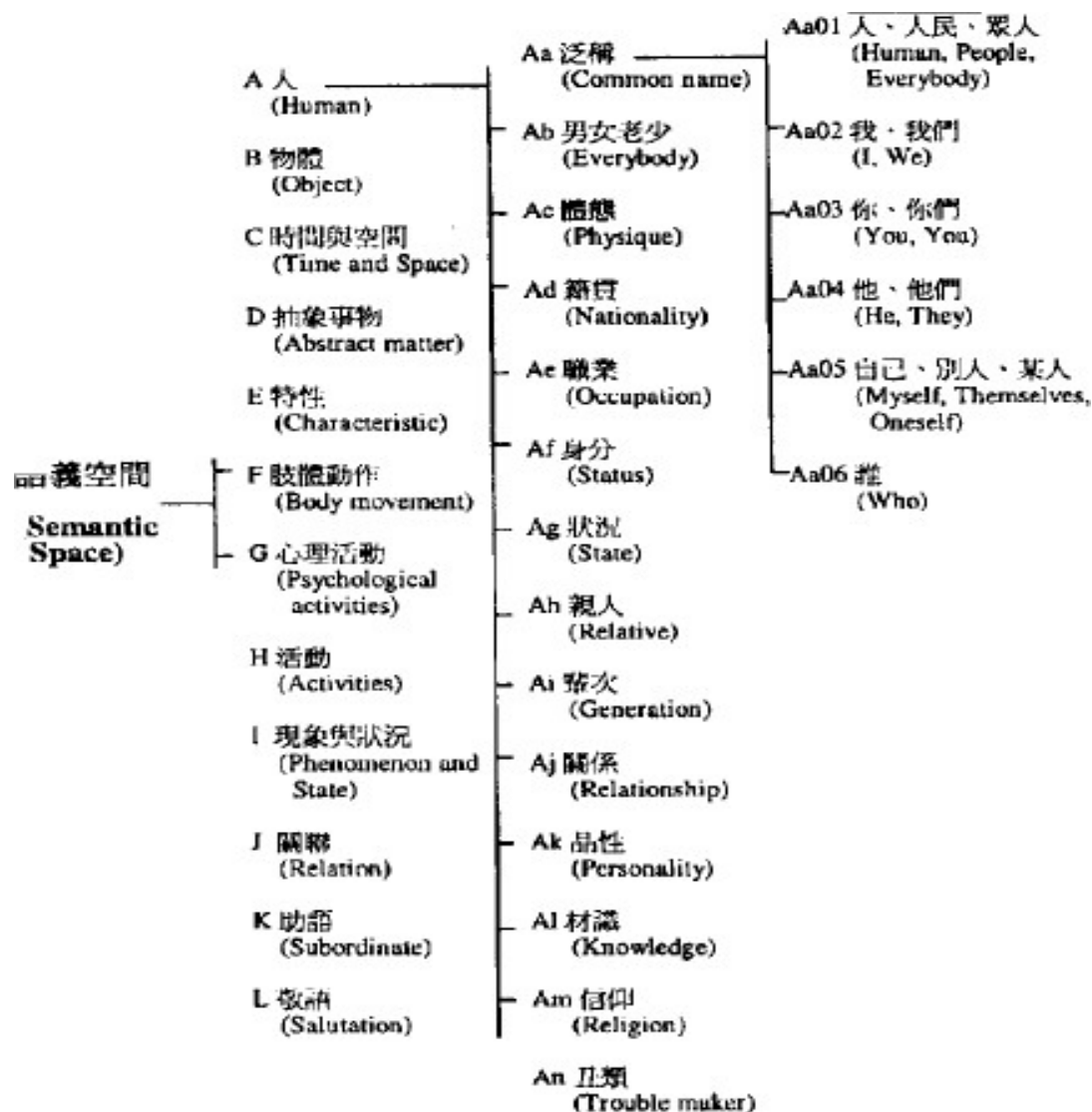


TongYiCiCiLin- CILIN

- A contemporary Chinese thesaurus
- Developed for translation and writing purposes
- CILIN contains approximately 63,600 entries for about 52,500 Chinese words
- Classifies commonly used Chinese words in a three-level hierarchical tree structure
- This hierarchical structure reflects the semantic relationship between words
- Defined by 12 major (top level), 95 medium (middle level), and 1428 minor (bottom level) semantic clusters
- Each minor semantic cluster consists of a set of words.
- Words under the same minor semantic cluster share the concept of this class



TongYiCiCiLin- CILIN





TongYiCiCiLin- CILIN

- CILIN is a valuable knowledge source for Chinese NLP
- Lua use the conventional simple co-occurrence approach for extracting the semantic knowledge from the CILIN
- Wong proposed an approach to extract partial semantic information from the CILIN
- New version of CILIN



HowNet 知网

- Developed by Dong Zhendong and Dong Qiang
 - <http://www.keenage.com/>
- HowNet is a extra-linguistic common-sense knowledge system for the computation of meaning in human language technology
- Unveils inter-concept relations and inter- attribute relations of concepts co-annoted in its Chinese-English lexicon



HowNet : Features

- The concept definition (DEF) in HowNet is based on sememes
 - Sememes are in a structured markup language.
 - 监狱 (prison)
 - DEF={InstitutePlace | 场所:domain={police | 警}, {detain | 扣住:location={~}, patient={human | 人:modifier={guilty | 有罪}}}, {punish | 处罚:location={~}, patient={human | 人:modifier={guilty | 有罪}}}}
 - “A prison is an institutional place, which is the location where guilty people are detained and punished. The place belongs to the domain of police and law”.



HowNet : Features

- HowNet constructs a **graph structure** of its knowledge base on the inter-concept relations and inter-attribute relations
 - Different from tree-structure lexical databases
 - Reveals concept relations within the same parts-of-speech (POS) categories and across POS categories
 - Especially the semantic-role relations between nouns and verbs.
- The representation is based on concepts denoted by words and expressions in both Chinese and English



HowNet: Sememes

- Hypothesizes that all concepts can be reduced to the relevant sememes
- Defines a close set of sememes from which an open set of concepts can be composed
- Sememes in HowNet are obtained by examining about 4,000 frequently used characters and their senses.
- This large-scale linguistic engineering process takes about three years of annotation and modification.
- 2,088 sememes in HowNet

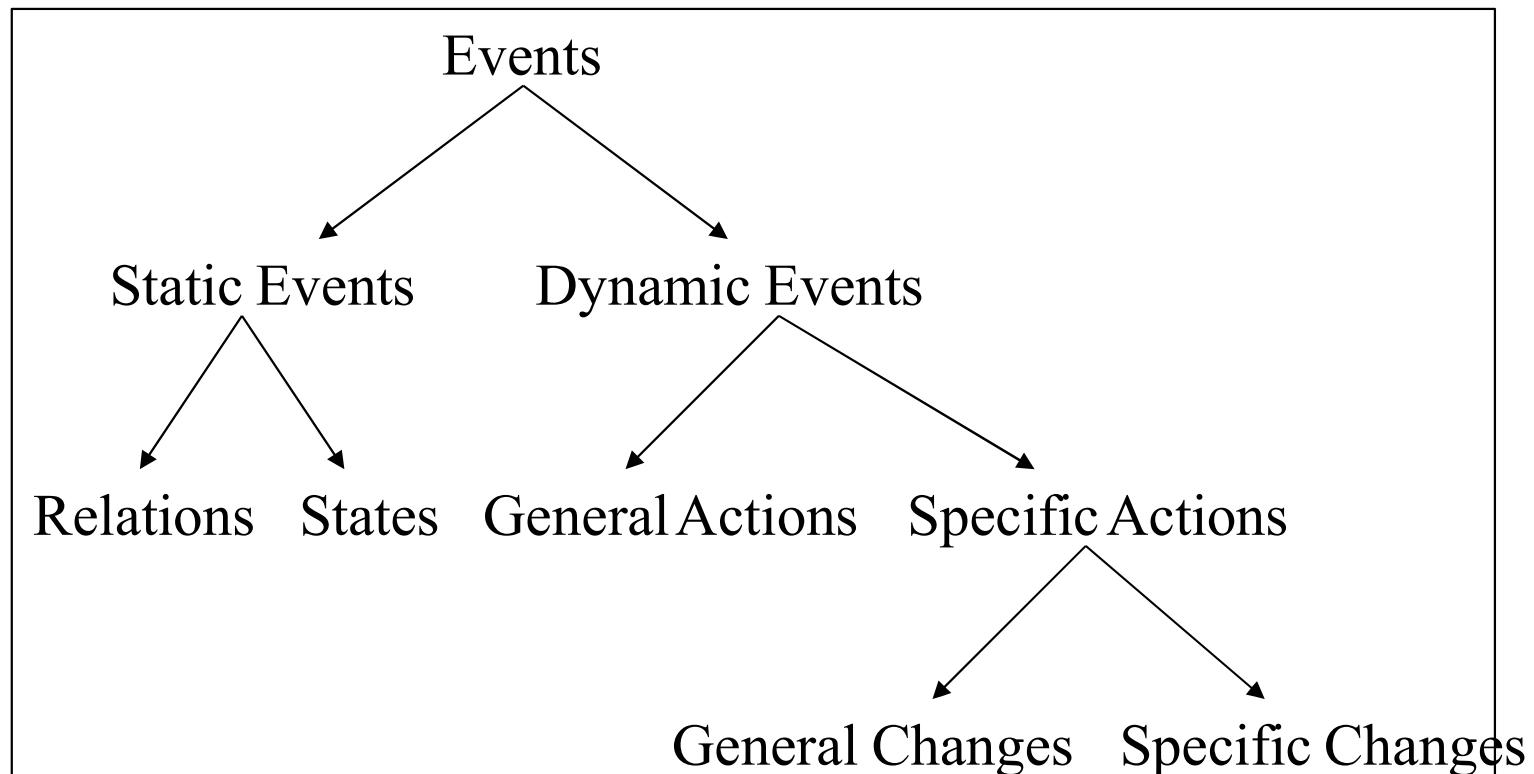


HowNet: Taxonomy

- Sememes are classified in a hierarchical structure, called taxonomy
- Taxonomy mainly provides the hypernym- hyponym relations of concepts
- 4 taxonomies in HowNet organizes its
 - taxonomy of events: 805 event sememes
 - taxonomy of entity, including things, parts, time and space: 152 entity sememes
 - taxonomy of attributes: 245 attribute sememes
 - taxonomy of attribute-values: 886 attribute- value sememes.

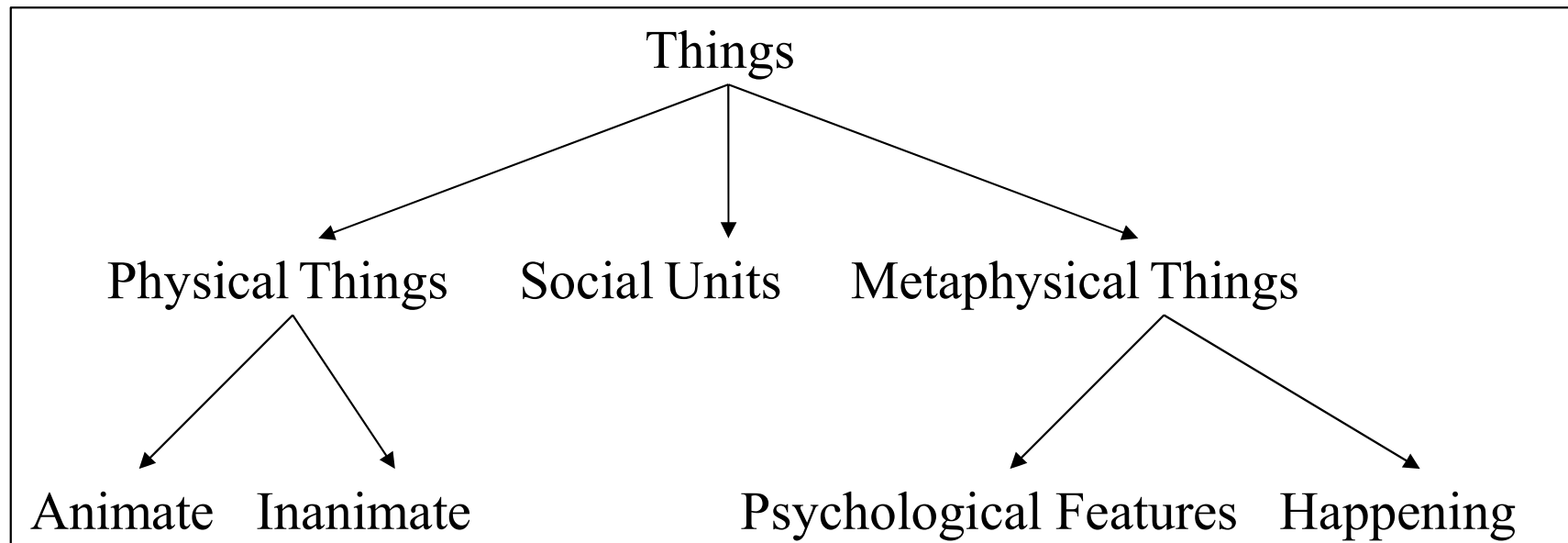


HowNet: Taxonomy - Event





HowNet: Taxonomy - Thing

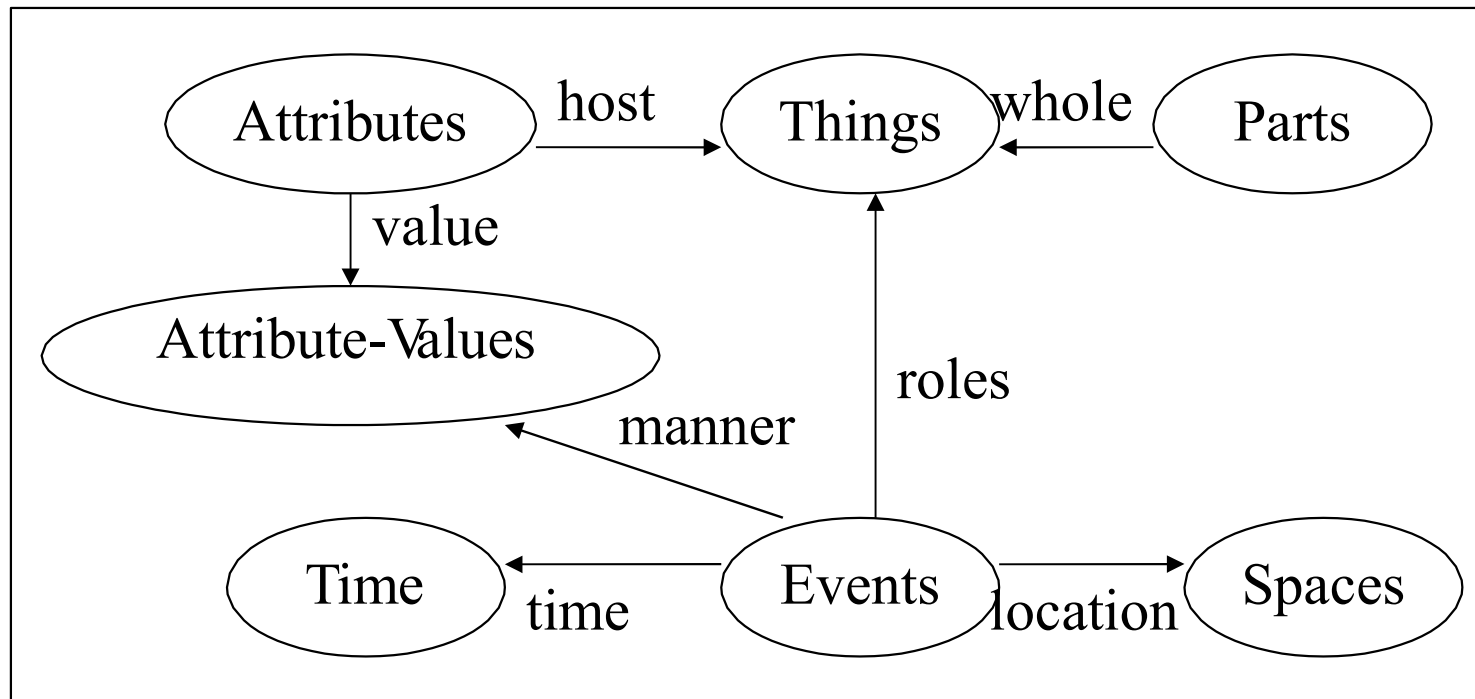




HowNet: Relations

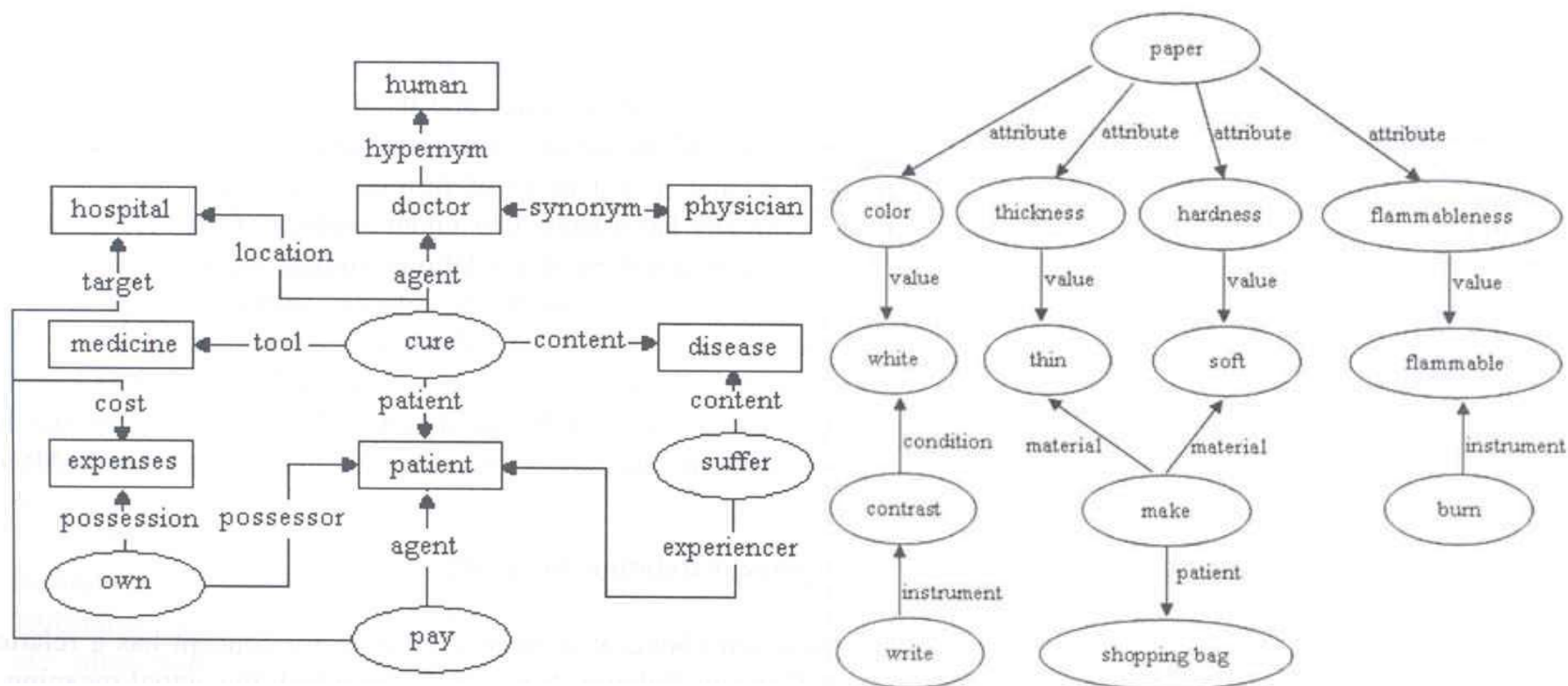
- Relations are the soul of HowNet.
- Classified into explicit relations and implicate relations
 - 11 types of explicit relation in HowNet.
 - (1) synonym (), (2) synclass, (3) antonym, (4) converse, (5) hypernym, (6) hyponym, (7) part- whole, (8) value-attribute, (9) attribute-host, (10) cognate role-frame, and (11) semantic roles-event.
 - the last four types are cross POS relations.

HowNet: Relation of Top-level Sememes



HowNet: Relations

- Divided into concept relation and attribute relation



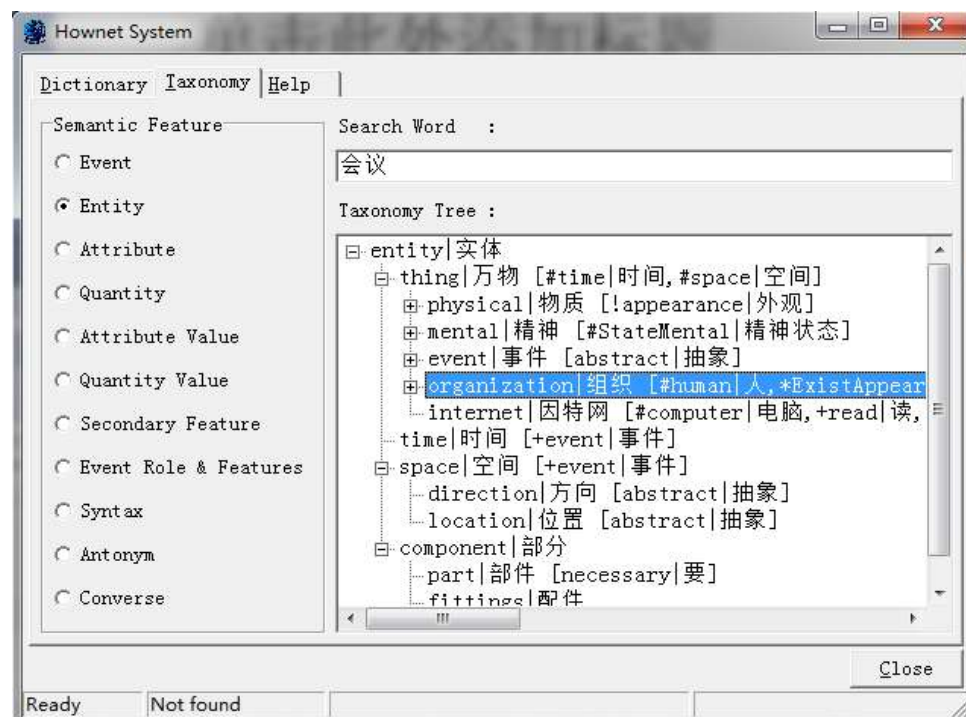
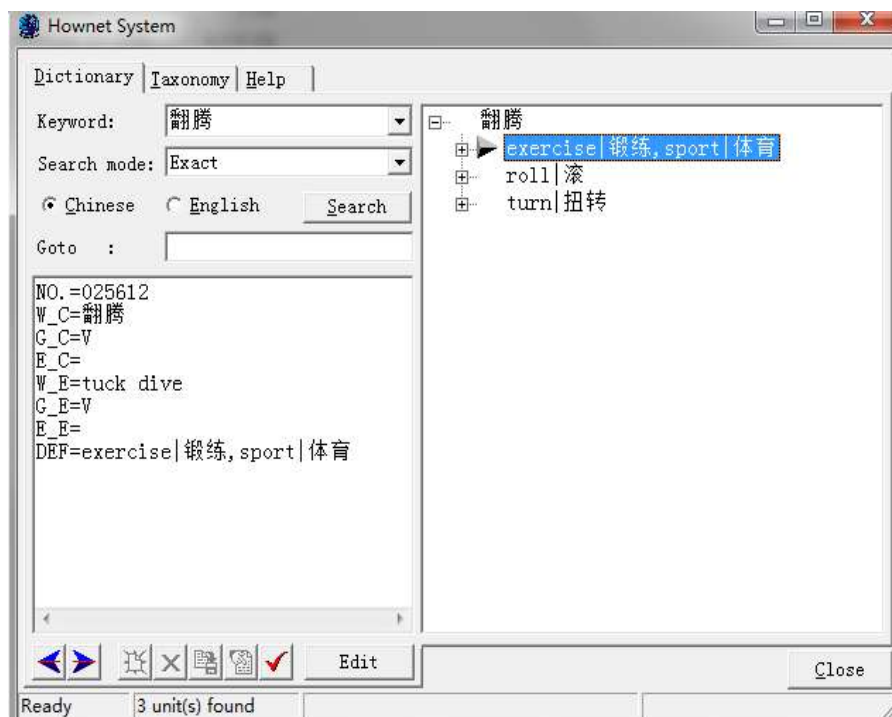


HowNet: Relations

- (a) 上下位关系（由概念的主要特征体现，请参看《知网管理工具》）
- (b) 同义关系（可通过《同义、反义以及对义组的形成》获得）
- (c) 反义关系（可通过《同义、反义以及对义组的形成》获得）
- (d) 对义关系（可通过《同义、反义以及对义组的形成》获得）
- (e) 部件-整体关系（由在整体前标注 % 体现，如“心”，“CPU”等）
- (f) 属性-宿主关系（由在宿主前标注 & 体现，如“颜色”，“速度”等）
- (g) 材料-成品关系（由在成品前标注 ? 体现，如“布”，“面粉”等）
- (h) 施事/经验者/关系主体-事件关系（由在事件前标注 * 体现，如“医生”，“雇主”等）
- (i) 受事/内容/领属物等-事件关系（由在事件前标注 \$ 体现，如“患者”，“雇员”等）
- (j) 工具-事件关系（由在事件前标注 * 体现，如“手表”，“计算机”等）
- (k) 场所-事件关系（由在事件前标注 @ 体现，如“银行”，“医院”等）
- (l) 时间-事件关系（由在事件前标注 @ 体现，如“假日”，“孕期”等）
- (m) 值-属性关系（直接标注无须借助标识符，如“蓝”，“慢”等）
- (n) 实体-值关系（直接标注无须借助标识符，如“矮子”，“傻瓜”等）
- (o) 事件-角色关系（由加角色名体现，如“购物”，“盗墓”等）
- (p) 相关关系（由在相关概念前标注 # 体现，如“谷物”，“煤田”等）



HowNet: Interface





HowNet: Word Similarity Estimation

- For two Chinese words W_1 and W_2 in HowNet, if W_1 has concept: $S_{11}, S_{12}, \dots, S_{1n}$, W_2 has concept: $S_{21}, S_{22}, \dots, S_{2m}$, the similarity between W_1 and W_2 is equals to the maximum similarity between two concepts

$$Sim(W_1, W_2) = \max_{i=1..n, j=1..m} Sim(S_{1i}, S_{2j})$$

HowNet: Word Similarity Computing



- Different from traditional semantic dictionary, each concept can **not** project to a concept tree node in HowNet, it is described by a series of primitive (义原).

similarity	function word concept (虚词概念)	notional word concept (实词概念)
function word concept	Similarity of corresponding syntactic primitive(句法义原) or relation primitive(关系义原)	0
notional word concept	0	sum of 4 parts: 1. 第一独立义原描述式 $Sim_1(S_1, S_2)$ 2. 其他独立义原描述式 $Sim_2(S_1, S_2)$ 3. 关系义原描述式 $Sim_3(S_1, S_2)$ 4. 符号义原描述式 $Sim_4(S_1, S_2)$

HowNet: Word Similarity Computing



- The semantic distance between two primitive is defined as:

$$Sim(p_1, p_2) = \frac{\alpha}{d + \alpha}$$

- d is the length of the path between p_1 and p_2 in the primitive hierarchy of HowNet.
 - α is an adjustable parameter.
- $Sim_1(S_1, S_2)$ can be calculated according to the above formula, the other three are different.

Applications of HowNet

- Word Representation Learning
 - [Niu et al. 2017] present that, word sememe information can improve word representation learning (WRL), which maps words into a low-dimensional semantic space and serves as a fundamental step for many NLP tasks.

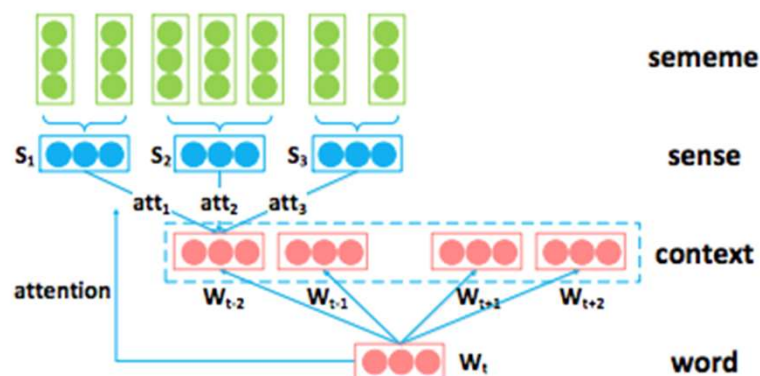


Figure 2: Sememe Attention over Context Model.

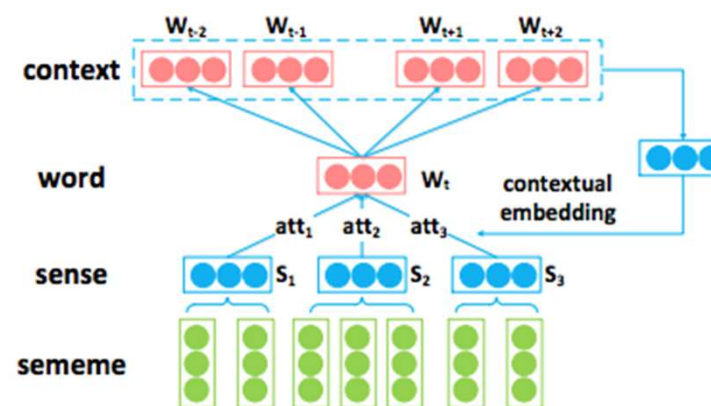


Figure 3: Sememe Attention over Target Model.

Applications of HowNet

- Event Detection
 - [Ding et al. 2019] propose the Trigger-aware Lattice Neural Network (TLNN).
 - For polysemous characters and words, this method model all senses of them with the help of HowNet, so as to alleviate the problem of ambiguous triggers.

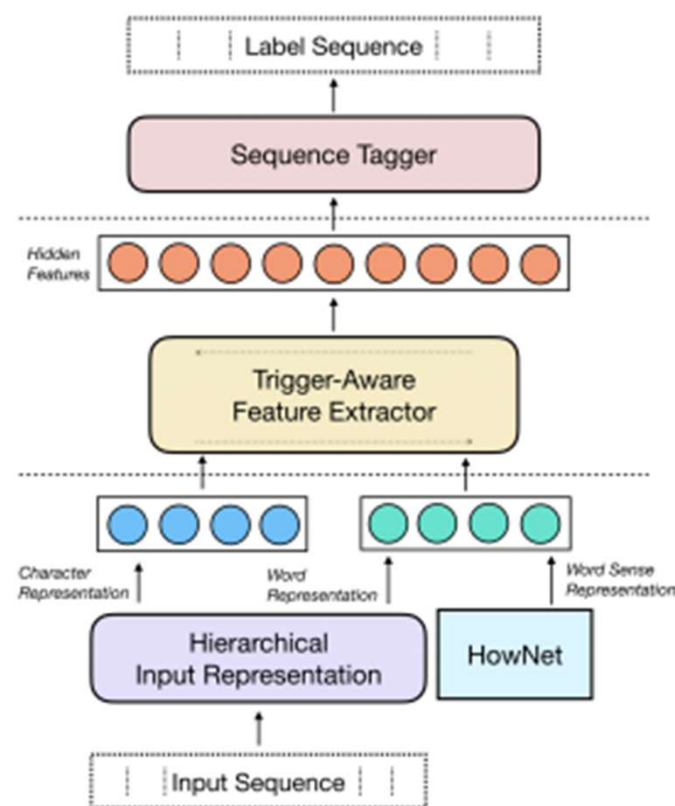


Figure 2: The architecture of TLNN.

CCD: Chinese Concept Dictionary



中文概念词典

- Developed in ICT, Peking University
- A bilingual Chinese-English WordNet-like Lexicon follow the framework of WordNet
- Compatible with WordNet structurally
- CCD records
 - synonyms, antonyms
 - hypernyms-hyponyms
 - meronyms-holonyms
 - collocations of words
 - major POS
 - nouns, verbs, adjectives and adverbs



CCD: Characteristics

- Nouns in CCD include
 - words of time, e.g. 下午 (afternoon),
 - location, e.g. 西部 (Western)
 - direction-position, e.g. 上 (up)
 - numerals, e.g. 甲 (first)
 - quantifiers, e.g. 批 (lots),
 - Pronouns
 - differentiation, e.g. 男/女 (male/female)
 - suffixes, e.g. 性 (gender), 器 (apparatus)
 - idioms, e.g. 八拜之交, 铜墙铁壁
 - semi-idioms (木头疙瘩、光杆司令)
 - abbreviated forms (政协)



-
- The intension and extension of verbs, adjectives and adverbs in CCD are different from those in WordNet
 - Semantically, concepts from different POSs may be related
 - 战争/n(battle) and 打仗 /v(fighting in a war) 战斗/v (fight) are associated
 - CCD alleviates this restriction in WordNet.
 - An attribute of association is appended to the feature structure of a concept.



-
- Relations described in CCD are much finer than WordNet
 - Synonyms relation is divided into equivalence and similarity
 - Antonyms relation is divided into essential antonyms and nonessential ones
 - Three new types of meronyms-holonyms relations are introduced
 - Examples of collocations are selected from real corpora with quantitative descriptions.



Nouns in CCD

- Synonymy
 - CCD records the word formation rule of the nouns if applicable.
 - attribute+part+noun->noun (长 *long*+鹿 *deer* + 颈 *neck* is a part of 鹿 *deer*->长颈鹿 *giraffe*)
 - CCD records the syntactic relations between the morphemes forming a noun.
 - 美人 *beautiful girl* has 美 *beautiful* and 人 *person*.
 - CCD records the refined noun categorization information
 - Abbreviations 北大 vs. 北京大学 Peking University
 - Regular and irregular words 教师 vs. 孩子王
 - Sentiment polarity 教师 neutral vs. 臭老九 negative



Nouns in CCD

- Antonymy

- Divided into essential / nonessential
- Word A and word B is named essential antonym if under one specific condition, A is the only antonym choice.
 - 丈夫 *husband*-妻子 *wife* is essential antonym in the relationship of husband-wife.
- Word A and word B are named nonessential antonyms if under one specific condition, A is not the only antonym choice.
 - under the relationship of persons with ages, 老人 *elder* and 孩子 *children*, 老人 *elder* and 年轻人 *young man*



- Hyponymy

- Hyponymy is transitive and asymmetrical

树 tree--

=>木本植物 woody plant--

=>维管植物 vascular plant

=>植物, 植物生命体 plant

=>生命形式, 生物体, 生物 life form

=>实体 entity

- Less than 10 levels



- Meronymy

- a component part of individual w' . 车轮—车
- a member of set (composed by individual w').
 - 树—森林
- the stuff that w' is made from. 钢—钢板
- a subarea of w' . 石家庄—河北
- event w is a part of event w' . 开幕式—会议
- time w is a part of time w' . 唐朝—古代

- Attribute



Verbs in CCD

- Synonymy
 - equivalent synonym/near synonyms
 - 死,逝世,去世,升天
 - Provide abbreviation information
 - 死亡/ 死亡
 - Provide regular/irregular use
 - 死去世逝世/老归西上西天
 - Provide sentiment polarity of verbs
 - 死亡死亡
 - 去世逝世过世老仙逝 Positive
 - 完蛋一命呜呼见阎王 Negative



-
- Antonymy
 - Hypernymy
 - Entailment
 - The temporal interval (or point) of verb $\bar{U}$ embeds in that of $\acute{U}$. 打鼾(snore) entails 睡觉(sleep)
 - The temporal interval (or point) of verb $\bar{U}$ is equal to that of $\acute{U}$. 跛行(limp) entails 行走
 - The temporal interval of verb $\bar{U}$ has nothing to do with that of $\acute{U}$. But inferring back from $\bar{U}$, 成功(succeed) entails 尝试(try)
 - The temporal interval of verb $\bar{U}$ has nothing to do with that of $\acute{U}$. But inferring back from $\bar{U}$, 给(give) entails 拥有(own)



- Sentence Frame
 - semantic roles of agent and patient

AGN Agent 施事	COS Course 经事
ESS Essive 当事	GOL Goal 向事
EXP Experience 感 事	SCP Scope 范 围
GEN Genitive 领事	RSN Reason 缘由
PAT Patient 受 事	INT Intention 意 图
IMP Impellee 致 事	TIM Time 时 间
RST Result 结 果	SPA Space 空间
CON Content 内容	MAN Manner 方式
BRL Belongings 属 事	INS Instrument 工具
PAR Part 分事	MAT Material 材 料
CAT Category 类事	QNT Quantity 数量
COM Comitative 涉事	DUR Duration 历时
SOR Source 源事	FRQ Frequency 频次



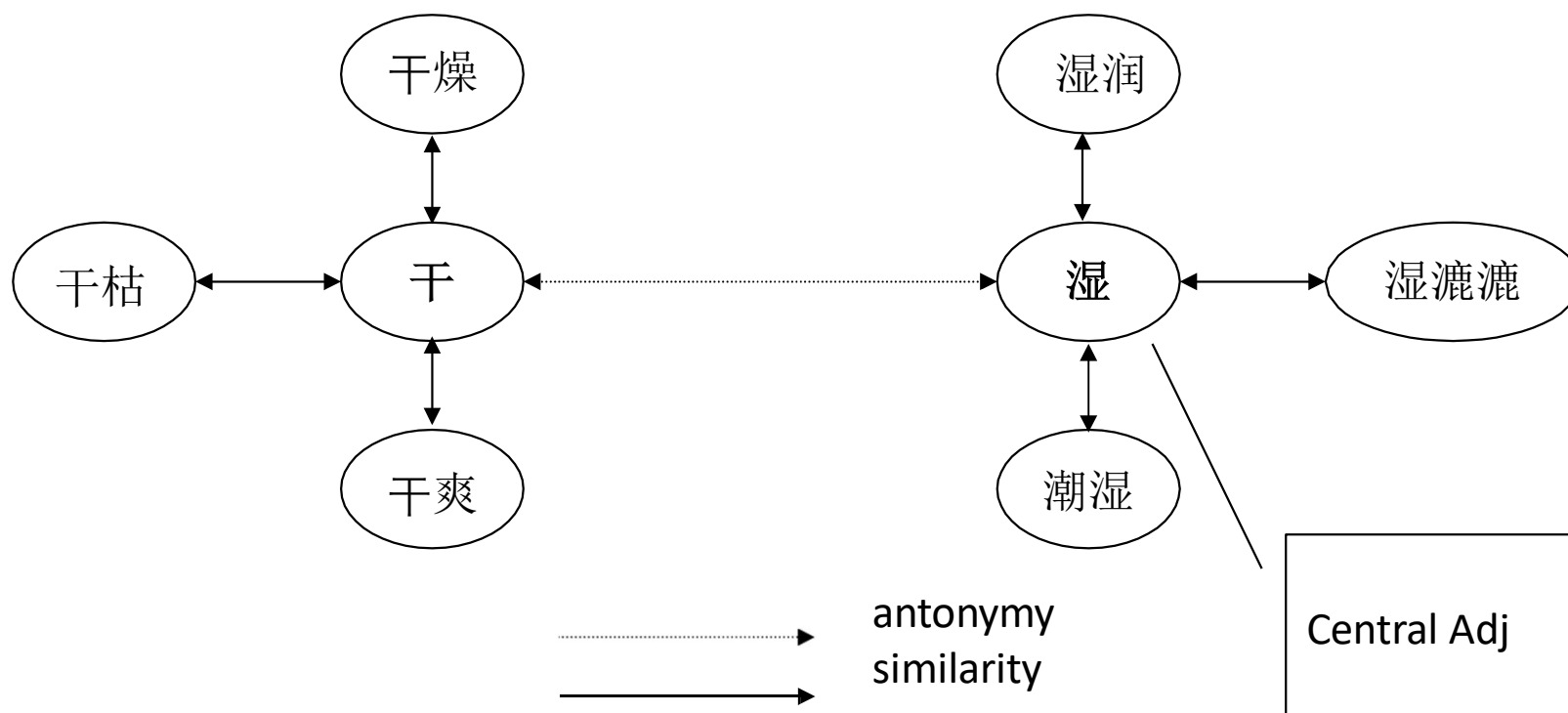
Adjectives in CCD

- Classifies Chinese adjectives into common adjectives and special adjectives
- CCD particularly emphasizes syntactic-semantic differences

Common Adjectives can be modified by 很 <i>very</i>	Special Adjectives Cannot be modified by 很 <i>very</i>		
大 小 高 低 重	Serve as adverbial modifier	Not serve as adverbial modifier	
	高速 大规 模 小 范围	Serve as attribute	Not serve as attribute
		男 女	红彤 彤 绿 油油



- Synonymy
 - 干(dry), {干, 干燥, 干枯, 干爽}
- Similarity
- Antonymy





Status of CCD

- CCD version 1.0 contains synsets and relations of
 - about 66,000 noun concepts
 - 12,000 verbal concepts
 - 21,000 adjectival and adverbial concepts
 - The annotated concepts can be aligned with about 100,000 concepts in English WordNet version 1.6.
- It plays an important role in the construction of Global WordNet.
- More information
http://icl.pku.edu.cn/icl_groups/ccd.asp.



Applications of Semantic Resources

- Word Sense Disambiguation
- Word Similarity Estimation
- Sentence Polarity Determination
- Ontology Construction
- Semantic Field Construction
- General Knowledge Reasoning
 - Transfer WordNet descripts to semantic network
 - hungry (饿) -- food (食物) -- refrigerator (冰箱)



Word Sense Disambiguation

- Background and Restrictions
- Automatic Word Sense Disambiguation
 - Knowledge-based Approach
 - Machine Learning based Approach
 - Supervised Method
 - Semi-supervised Method
 - Unsupervised Method
 - Hybrid Approach
- Evaluations



Examples (Yarowsky, 1995)

plant	living/factory
tank	vehicle/container
poach	steal/boil
palm	tree/hand
bass	fish/music
motion	legal/physical
crane	bird/machine



Harder Cases

- WordNet: Senses of “Line”

- [S](#): (n) **line** (a formation of people or things one beside another) *"the line of soldiers advanced with their bayonets fixed"; "they were arrayed in line of battle"; "the cast stood in line for the curtain call"*
- [S](#): (n) **line** (a mark that is long relative to its width) *"He drew a line on the chart"*
- [S](#): (n) **line** (a formation of people or things one behind another) *"the line stretched clear around the corner"; "you must wait in a long line at the checkout counter"*
- [S](#): (n) **line** (a length (straight or curved) without breadth or thickness; the trace of a moving point)
- [S](#): (n) **line** (text consisting of a row of words written across a page or computer screen) *"the letter consisted of three short lines"; "there are six lines in every stanza"*
- [S](#): (n) **line** (a single frequency (or very narrow band) of radiation in a spectrum)
- [S](#): (n) **line** (a fortified position (especially one marking the most forward position of troops)) *"they attacked the enemy's line"*
- [S](#): (n) [argumentation](#), [logical argument](#), [argument](#), [line of reasoning](#), **line** (a course of reasoning aimed at demonstrating a truth or falsehood; the methodical process of logical reasoning) *"I can't follow your line of reasoning"*
- [S](#): (n) [cable](#), **line**, [transmission line](#) (a conductor for transmitting electrical or optical signals or electric power)
- [S](#): (n) [course](#), **line** (a connected series of events or actions or developments) *"the government took a firm course"; "historians can only point out those lines for which evidence is available"*
- [S](#): (n) **line** (a spatial location defined by a real or imaginary unidimensional extent)
- [S](#): (n) [wrinkle](#), [furrow](#), [crease](#), [crinkle](#), [seam](#), **line** (a slight depression or fold in the smoothness of a surface) *"his face has many lines"; "ironing gets rid of most wrinkles"*
- [S](#): (n) [pipeline](#), **line** (a pipe used to transport liquids or gases) *"a pipeline runs from the wells to the seaport"*
- [S](#): (n) **line**, [railway line](#), [rail line](#) (the road consisting of railroad track and roadbed)



WSD: Motivation

- One of the central challenges in NLP.
- Ubiquitous across all languages.
- Needed in:
 - **Machine Translation:** For correct lexical choice.
 - **Information Retrieval:** Resolving ambiguity in queries.
 - **Information Extraction:** For accurate analysis of text.
- Computationally determining which sense of a word is activated by its use in a particular context.
 - E.g. I am going to withdraw money from the **bank**.
- A classification problem:
 - Senses → Classes
 - Context → Evidence



Automatic WSD Approaches

- **Knowledge Based Approach**
 - WSD using Selectional Preferences (or restrictions)
 - Overlap Based Approaches
- **Machine Learning Based Approach**
 - Supervised Approaches
 - Semi-supervised Algorithms
 - Unsupervised Algorithms
- **Hybrid Approach**
- **Neural Approach**



Overlap based Approach

- Require a ***Machine Readable Dictionary***
- Find the overlap between the features of different senses of target word (**sense bag**) and features of the words in its context (**context bag**).
- These features could be sense definitions, example sentences, hypernyms etc.
- The features could also be given weights
- The sense which has the maximum overlap is selected as the contextually appropriate one



LESK'S Algorithm

Sense Bag: contains the words in the definition of a candidate sense of the ambiguous word.

Context Bag: contains the words in the definition of each sense of each context word.

E.g. "On burning **coal** we get **ash**."

Ash

○ Sense 1

Trees of the olive family with pinnate leaves, thin furrowed bark and gray branches.

○ Sense 2

The **solid** residue left when **combustible** material is thoroughly **burned** or oxidized.

○ Sense 3

To convert into ash

Coal

○ Sense 1

A piece of glowing carbon or **burnt** wood.

○ Sense 2

charcoal.

○ Sense 3

A black **solid combustible** substance formed by the partial decomposition of vegetable matter without free access to air and under the influence of moisture and often increased pressure and temperature that is widely used as a fuel for **burning**

In this case Sense 2 of ash would be the winner sense.

WALKER'S Thesaurus Based Algorithm



- **Step 1:** For each sense of the target word find the thesaurus category to which that sense belongs.
- **Step 2:** Calculate the score for each sense by using the context words. A context words will add 1 to the score of the sense if the thesaurus category of the word matches that of the sense.
 - E.g. The money in this **bank** fetches an interest of 8% per annum
 - Target word: **bank**
 - Clue words from the context: **money**, **interest**, **annum**, **fetch**

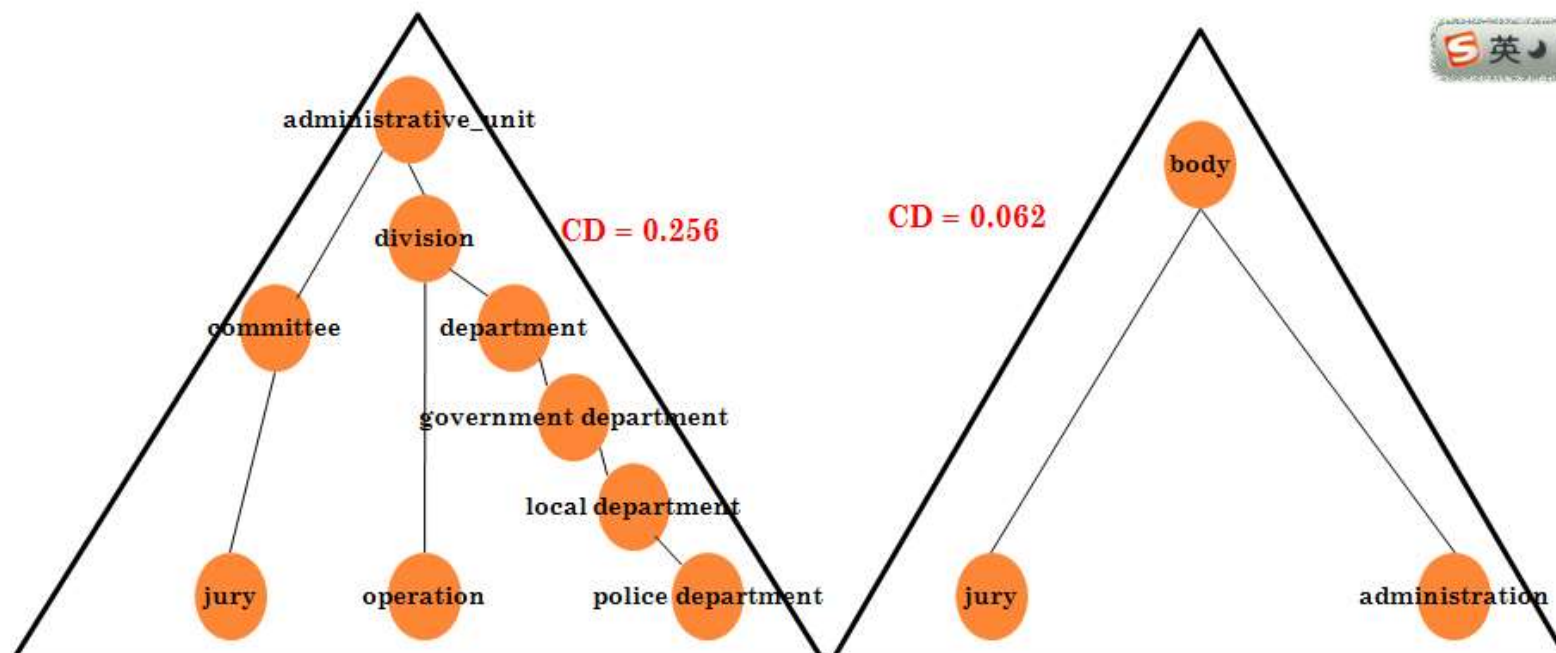
	Sense1: Finance	Sense2: Location
Money	+1	0
Interest	+1	0
Fetch	0	0
Annum	+1	0
Total	3	0

Context words add 1 to the sense when the topic of the word matches that of the sense



Conceptual Density based WSD

- Select a sense based on the relatedness of that word-sense to the context.
- Relatedness is measured in terms of conceptual distance
 - *(i.e. how close the concept represented by the **word** and the concept represented by its **context words** are)*
- This approach uses a structured hierarchical semantic net (WordNet) for finding the conceptual distance.
- Smaller the conceptual distance higher will be the conceptual density.
 - *(i.e. if all words in the context are strong indicators of a particular concept then that concept will have a higher density.)*



The jury(2) praised the administration(3) and operation (8) of Atlanta Police Department(1)

Step 1: Make a lattice of the nouns in the context, their senses and hypernyms.

Step 3: The concept with highest CD is selected.

Step 2: Compute the conceptual density of resultant concepts

Step 4: Select the senses below the selected concept as the correct sense for the respective words.

Knowledge-based Approach: Conclusions



- Drawbacks of WSD using Selectional Restrictions
 - Needs exhaustive Knowledge Base.
- Drawbacks of Overlap based approaches
 - Dictionary definitions are generally very small.
 - Dictionary entries rarely take into account the distributional constraints of different word senses (e.g. selectional preferences, kinds of prepositions, etc. → **cigarette** and **ash** never co-occur in a dictionary).
 - Suffer from the problem of sparse match.
 - Proper nouns are not present in a MRD. Hence these approaches fail to capture the strong clues provided by proper nouns.

*E.g. “**Sachin Tendulkar**” will be a strong indicator of the category “**sports**”.*

***Sachin Tendulkar** plays **cricket**.*



Automatic WSD Approaches

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 - Unsupervised Algorithms
- **Hybrid Approach**
- **Neural Approach**
- Machine learning algorithm is applied to process WSD as a classification



Two Assumptions (Yarowsky 1995)

- **One Sense Per Collocation**

- nearby words provide strong and consistent clues to the sense of a target word, conditional on relative distance, order and syntactic relationship
- 95-96% applicable

- **One Sense Per Discourse**

- The sense of a target word is highly consistent within any given document.
- True for topic dependent words
- Not true for verbs
- Krovetz (1998): not true with respect to fine-grained senses: (e.g., language/people)

Intuition of distributional word similarity



- Nida example:
 - A bottle of **tesgüinois** on the table!
 - Everybody likes **tesgüino**!
 - Tesgüino** makes you drunk!
 - We make **tesgüino** out of corn.
- From context words, humans guess tesgüino means an alcoholic beverage like beer.
- Intuition for algorithm:
 - Two words are similar if they have similar word contexts.



Reminder: Term document matrix

- Each cell: count of term t in a document d : $t_{t,d}$:
 - Each document is a count vector in $\mathbb{N}^V$: a column below

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	1	8	15
soldier	2	2	12	36
fool	37	58	1	5
clown	6	117	0	0

- Two documents are similar if their vectors are similar

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	1	8	15
soldier	2	2	12	36
fool	37	58	1	5
clown	6	117	0	0



Reminder: Term document matrix

- Each word is a **count vector** in $\mathbb{N}^D$: a row below

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	1	8	15
soldier	2	2	12	36
fool	37	58	1	5
clown	6	117	0	0

- Two **words** are similar if their vectors are similar

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	1	8	15
soldier	2	2	12	36
fool	37	58	1	5
clown	6	117	0	0



Reminder: Term document matrix

- Instead of using entire documents, use smaller contexts
 - Paragraph
 - Window of 10 words
- A word is now defined by a vector over counts of context words
- For the term-document matrix
 - We used **tf-idf** instead of raw term counts
 - **Positive Pointwise Mutual Information (PPMI)** is common (**PPMI** Replace all **PMI** values less than 0 with zero)



Supervised Methods for WSD

- (Firth) "You shall know the word by the company it keeps"
- Supervised sense disambiguation is very successful
- However, it requires a lot of data
- A supervised method : Decision List



WSD: Naïve Bayes

$$\hat{s} = \operatorname{argmax}_{s \in \text{senses}} \Pr(s|V_w)$$

where V_w is a feature vector consisting of:

- POS of w
- Semantic & Syntactic features of w
- Collocation vector (set of words around it) $\rightarrow$ typically consists of next word(+1), next-to-next word(+2), -2, -1 & their POS's
- Co-occurrence vector (number of times w occurs in bag of words around it)

Applying Bayes rule and naive independence assumption

$$\hat{s} = \operatorname{argmax}_{s \in \text{senses}} \Pr(s) \cdot \prod_{i=1}^n \Pr(V_w^i | s)$$



Contextual Features in WSD

- Word found in $\pm k$ word window
- Word immediately to the right (+1 W)
- Word immediately to the left (-1 W)
- Pairs of words at offsets -2 and -1
- Pair of words at offsets -1 and +1
- Pair of words at offsets +1 and +2
- Part of speech of contextual words
-
- Some features are represented by their classes (WEEKDAY, MONTH)



Example

The ocean reflects the color of the sky, but even on cloudless days the color of the ocean is not a consistent blue. Phytoplankton, microscopic **plant** life that floats freely in the lighted surface waters, may alter the color of the water. When a great number of organisms are concentrated in an area, ...

w_{-1} = microscopic $t_1 = JJ$

w_{+1} = life $t_{+1} = NN$

$w_{-2}, w_{-1} = (\text{Phytoplankton, microscopic}) \dots$

$w_{-1}, w_{+1} = (\text{microscopic, life})$

word-within-k=ocean

word-within-k=reflects

...



Decision Lists

- Based on ‘One sense per collocation’ property.
 - Nearby words provide strong and consistent clues as to the sense of a target word.
- Collect a large set of collocations for the ambiguous word.
- Calculate word-sense probability distributions for all such collocations.
- Calculate the log-likelihood ratio

$$\text{Log}\left(\frac{\text{Pr}(\text{Sense-A} | \text{Collocation}_i)}{\text{Pr}(\text{Sense-B} | \text{Collocation}_i)}\right)$$

Assuming there are only two senses for the word. Of course, this can easily be extended to ‘k’ senses.

- Higher log-likelihood = more predictive evidence
- Collocations are ordered in a **decision list**, with most predictive collocations ranked highest.



Creating Decision Lists

- For each feature, find
$$\text{sense}(\text{feature}) = \underset{\text{senses}}{\text{argmax}} P(\text{sense}|\text{feature})$$
e.g., $\text{sense}(w_{+1}=\text{life})=\text{sense}_1$
- Create a rule $\text{feature} \rightarrow \text{sense}(\text{feature})$ with weight $P(\text{sense}(\text{feature})|\text{feature})$

Rule	Weight
$w_{+1} = \text{life} \rightarrow \text{plant}_1$	0.99
$w_{+1} = \text{work} \rightarrow \text{plant}_2$	0.93



Creating Decision Lists

- Create a list of rules sorted by strength

Rule	Weight
$w_{+1} = life \rightarrow plant_1$	0.99
$w_{-1} = modern \rightarrow plant_2$	0.98
$w_{+1} = work \rightarrow plant_2$	0.975
word-within-k= $life \rightarrow plant_1$	0.95
$w_{-1} = assembly \rightarrow plant_2$	0.94

- To apply the decision list: take the first rule in the list which applies to an example



Applying Decision Lists

The ocean reflects the color of the sky, but even on cloudless days the color of the ocean is not a consistent blue. Phytoplankton, microscopic **plant** life that floats freely in the lighted surface waters, may alter the color of the water. When a great number of organisms are concentrated in an area, ...

Feature	Sense	Strength
w_{-1} = microscopic	1	0.95
w_{+1} = life	1	0.99
w_{-2}, w_{-1} =	N/A	
word-within-k=reflects	2	0.65
...		

- N/A $\rightarrow$ feature has not seen in training data
 w_{+1} = life Sense_1 is chosen



Experimental Results

- (Yarowsky, 1995)
- Accuracy of 95% on binary WSD

plant	living/factory
tank	vehicle/container
poach	steal/boil
palm	tree/hand



WSD based on SVMs

- SVM is a binary classifier which finds a hyperplane with the largest margin that separates training examples
- A separate classifier is built for each sense of the word
- **Training Phase:** Using a tagged corpus, for every sense of the word, a SVM is trained
 - POS of **w** as well as POS of neighboring words.
 - Local collocations - Co-occurrence vector
 - Features based on syntactic relations (e.g. headword, POS of headword, voice of head word etc.)
- **Testing Phase:** Given a test sentence, a test example is constructed using the above features and fed as input to each binary classifier.
- The correct sense is selected based on the label returned by each classifier.



Exemplar Based WSD (k-NN)

- An exemplar based classifier is constructed for each word to be disambiguated.

Step1: From each *sense marked sentence* containing the ambiguous word , a training example is constructed using:

- POS of **w** as well as POS of neighboring words. - Local collocations
- Co-occurrence vector - Morphological features
- Subject-verb syntactic dependencies

Step2: Given a test sentence containing the ambiguous word, a test example is similarly constructed.

Step3: The test example is then compared to all training examples and the k-closest training examples are selected.

Step4: The sense which is most prevalent amongst these “k” examples is then selected as the correct sense.



Beyond Supervised Methods

- If you want to be able to do WSD in the large, you need to be able to disambiguate all words in a text
- It is hard to get large amount of annotated data for every word in text
 - Use existing manually tagged data (SENSEVAL-2, 5000 words from Penn Treebank)
 - Use parallel bilingual data
 - Check OpenMind Word Expert project
<http://teach-computers.org/word-expert.html>



Supervised Methods: Comparisons

Approach	Average Precision	Average Recall	Corpus	Average Baseline Accuracy
Naïve Bayes	64.13%	Not reported	Senseval3 – All Words Task	60.90%
Decision Lists	96%	Not applicable	Tested on a set of 12 highly polysemous English words	63.9%
Exemplar Based disambiguation (k-NN)	68.6%	Not reported	WSJ6 containing 191 content words	63.7%
SVM	72.4%	72.4%	Senseval 3 – Lexical sample task (Used for disambiguation of 57 words)	55.2%
Perceptron trained HMM	67.60	73.74%	Senseval3 – All Words Task	60.90%



Supervised Methods: Conclusions

- General Comments

- Use corpus evidence instead of relying of dictionary defined senses.
- Can capture important clues provided by proper nouns because proper nouns do appear in a corpus.

- Naïve Bayes

- Suffers from data sparseness.
- Since the scores are a product of probabilities, some weak features might pull down the overall score for a sense.
- A large number of parameters need to be trained.

- Decision Lists

- A word-specific classifier. A separate classifier needs to be trained for each word.
- Uses the single most predictive feature which eliminates the drawback of Naïve Bayes.



Supervised Methods: Conclusions

- Exemplar Based K-NN

- A word-specific classifier.
- Will not work for unknown words which do not appear in the corpus.
- Uses a diverse set of features (including morphological and noun-subject-verb pairs)

- SVM

- A word-sense specific classifier.
- Gives the highest improvement over the baseline accuracy.
- Uses a diverse set of features.

- HMM

- Significant in lieu of the fact that a fine distinction between the various senses of a word is not needed in tasks like MT.
- A broad coverage classifier as the same knowledge sources can be used for all words belonging to super sense.
- Even though the polysemy was reduced significantly there was not a comparable significant improvement in the performance.



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Semi-supervised Decision List Algorithm



- Based on Yarowsky's supervised algorithm that uses *Decision Lists*.
- Collecting seed examples
 - Goal: start with a small subset of the training data being labeled
 - Label a number of training examples by hand
 - Pick a single feature for each class by hand
 - Use words in dictionary definitions
 - a vegetable organism, ready for **planting**
 - equipment, machinery, apparatus, for industrial **plant**

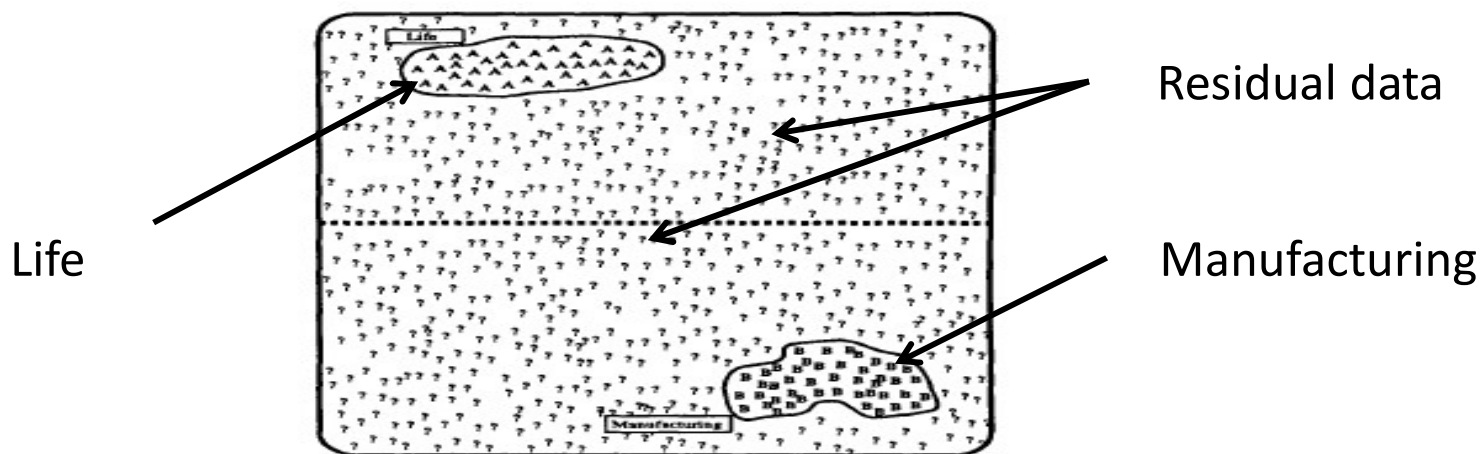


Collecting Seed Examples:

- For the “**plant**” sense distinction, initial seeds are “word-within-k=life” and “word-within-k=manufacturing”
- Partition the unlabeled data into three sets:
 - 82 examples labeled with “life” sense
 - 106 examples labeled with “manufacturing” sense
 - 7350 unlabeled examples

Iterative Bootstrapping – Step 1

- Identify all contexts in which the polysemous word occurs.
- For each possible sense use seed collocations to identify a relatively small number of training examples representative of that sense.



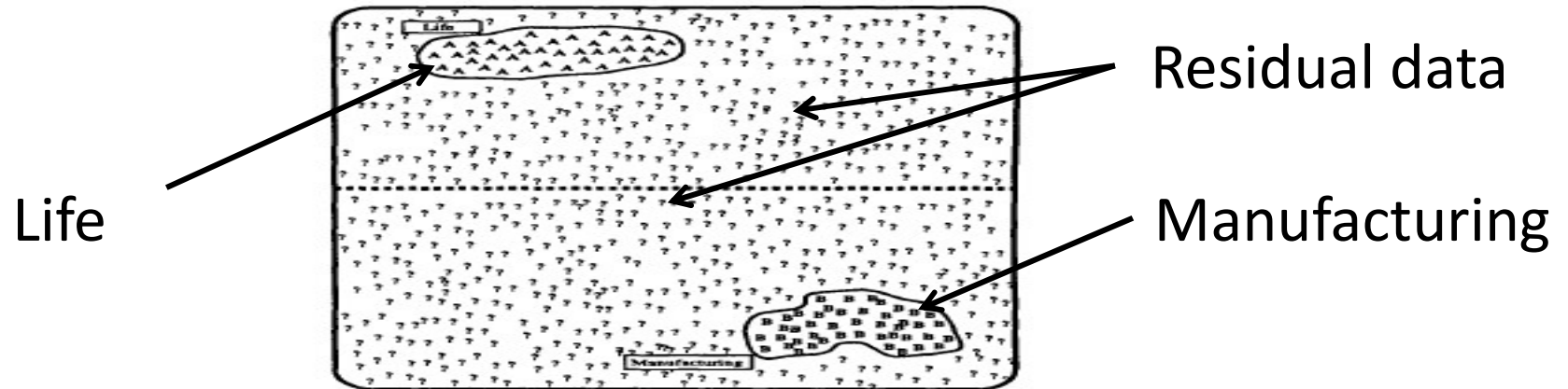
- Seed collocation should accurately distinguish the senses.



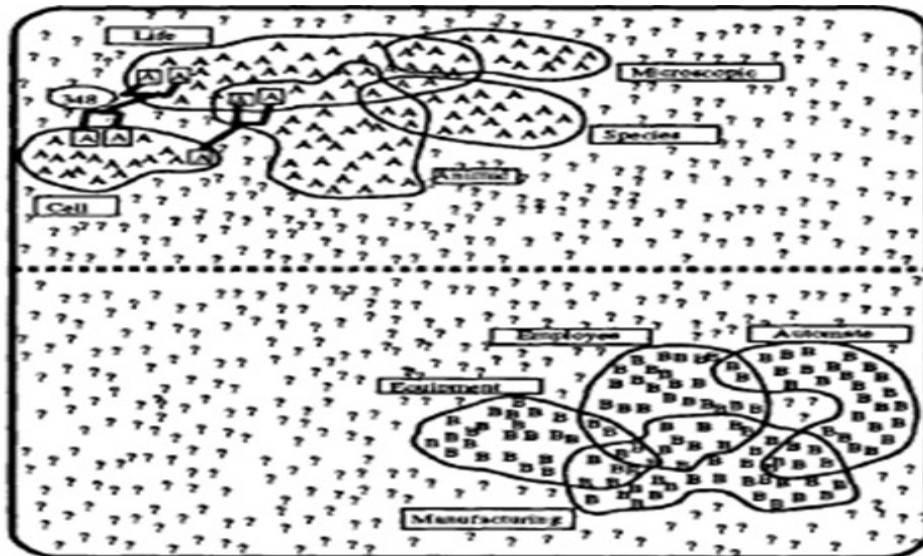
Iterative Bootstrapping – Step 2

- Train the Decision List algorithm on the seed data.
- Classify the entire sample set using the trained classifier.
- Create new seed data by adding those members which are tagged as Sense-A or Sense-B with high probability.
- Retrain the classifier using the new seed data.
- These additions will contribute new collocations that are reliably indicative of the 2 senses.

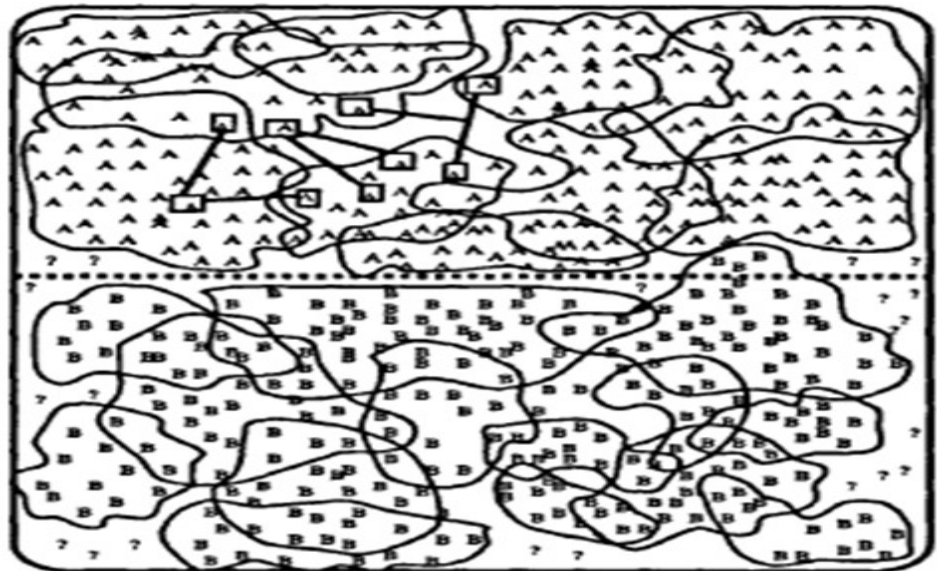
Initialization, Progress and Convergence



Seed set grows



Stop when residual set





Semi-Supervised Methods

Approach	Average Precision	Corpus	Average Baseline Accuracy
Supervised Decision Lists	96.1%	Tested on a set of 12 highly polysemous English words	63.9%
Semi-Supervised Decision Lists	96.1%	Tested on a set of 12 highly polysemous English words	63.9%

- Works at par with its supervised version even though it needs significantly less amount of tagged data.
- Has all the advantages and disadvantages of its supervised version.

WSD: Unsupervised Approach Using ROGET'S Thesaurus Categories



- Based on three observations:
 - Different conceptual classes of words tend to appear in recognizably different contexts.
 - Different word senses belong to different conceptual classes (E.g. crane 吊车/ 鹤)
 - A context based discriminator for the conceptual classes can serve as a context based discriminator for the members of those classes.
- Identify salient words in the collective context of the thesaurus category and weigh appropriately.

WSD: Unsupervised Approach Using ROGET'S Thesaurus Categories



$$\text{Weight}(\text{word}) = \text{Salience}(\text{Word}) = \frac{\text{Pr}(w|RCat)}{\text{Pr}(w)}$$

ANIMAL/INSECT

species (2.3), family(1.7), bird(2.6), fish(2.4), egg(2.2),
coat(2.5), female(2.0), eat (2.2), nest(2.5), wild

TOOLS/MACHINERY

tool (3.1), machine(2.7), engine(2.6), blade(3.8),
cut(2.2), saw(2.5), lever(2.0), wheel (2.2), piston(2.5)

WSD: Unsupervised Approach Using ROGET'S Thesaurus Categories



- Predict the appropriate category for an ambiguous word using the weights of words in its context:

$$\operatorname{argmax}_{w \text{ in context}} \log \left(\frac{\Pr(w|Rcat) * \Pr(Rcat)}{\Pr(w)} \right)$$

...lift water and to grind grain. Treadmills attached to *cranes* were used to lift heavy objects from Roman times,

TOOLS/MACHINE	Weight	ANIMAL/INSECT	Weight
lift	2.44	Water	0.76
grain	1.68		
used	1.32		
heavy	1.28		
Treadmills	1.16		
attached	0.58		
grind	0.29		
Water	0.11		
TOTAL	11.30	TOTAL	0.76



LIN'S Approach

Two different words are likely to have similar meanings if they occur in identical local contexts.

E.g. The **facility** will **employ** 500 new employees.

Senses of facility

- **installation**
- proficiency
- adeptness
- readiness
- toilet/bathroom

In this case **Sense 1** of installation would be the winner **sense**.

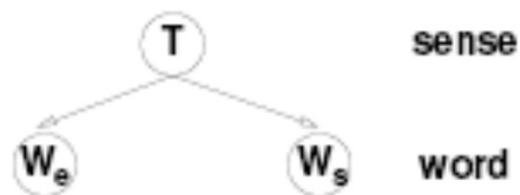
Subjects of “employ”

Word	Freq	Log Likelihood
ORG	64	50.4
Plant	14	31.0
Company	27	28.6
Industry	9	14.6
Unit	9	9.32
Aerospace	2	5.81
Memory device	1	5.79
Pilot	2	5.37



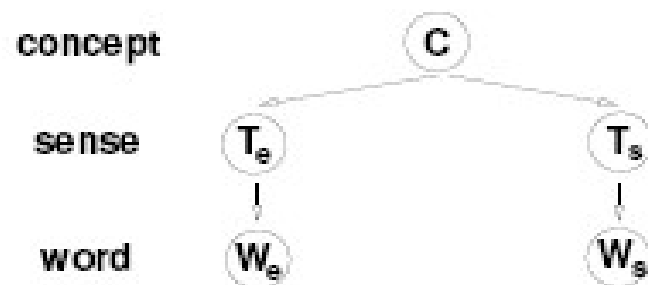
WSD: Using Bilingual Corpora

- A word having multiple senses in one language will have distinct translations
- The translations are considered as contextual indicators of the sense
- **Sense Model**



$$P(W_e, W_s, T) = P(T) \cdot P(W_e|T) \cdot P(W_s|T) \quad (5.1)$$

- **Concept Model**



$$P(W_e, W_s, T_e, T_s, C) = P(C) \cdot P(T_e|C) \cdot P(T_s|C) \cdot P(W_e|T_e) \cdot P(W_s|T_s) \quad (5.2)$$

WSD: Hybrid Approach

An Iterative Approach



- Uses semantic relations (synonymy and hypernymy) from WordNet.
- Extracts collocational and contextual information from WordNet (gloss) and a small tagged data.
- Monosemic words in the context serve as a seed set of disambiguated words.
- In each iteration new words are disambiguated based on their semantic distance from already disambiguated words.
- Precision 92.2% Recall 55%



WSD: EM algorithm using HowNet to exploit features

Step1: Feature Selection.

- Morphology: “广阔”and “辽阔”
“严肃”and “严厉”
“红火”and “火爆”

Use HowNet to get the core characters of a word:

被动 → 被
安全 → 安
高大 → 高 and 大



WSD: EM algorithm using HowNet to exploit features

- Use HowNet to get the core characters of a word.
 - Group the words: the words with the same characters are assigned to the same class.
 - Search in the HowNet synset description, remove the words which have no character in the description.

词语在 HowNet 中的义原描述

词语	HowNet 的义原描述
安全	aValue 属性值, circumstances 境况, safe 安, desired 良
被动	aValue 属性值, behavior 举止, passive 被
便捷	aValue 属性值, behavior 举止, convenient 便, desired 良 aValue 属性值, behavior 举止, nimble 捷, desired 良



WSD: EM algorithm using HowNet to exploit features

- Collocation: “健康发展”vs. “健康成长”
- Attribute-host relation(属性--宿主关系):
“像百合一样清纯”vs. “如雨后空气般清纯”

Step2: Word clustering using EM algorithm.

Clustering Step: Grouping all words into several clusters according to their sense word features. The word clusters corresponding to senses are obtained.

Sense Disambiguation: Calculate the association between each word and different word clusters. Stronger associations indicates that the words have higher probability to describe the sense. The top words are selected as the sense word features.

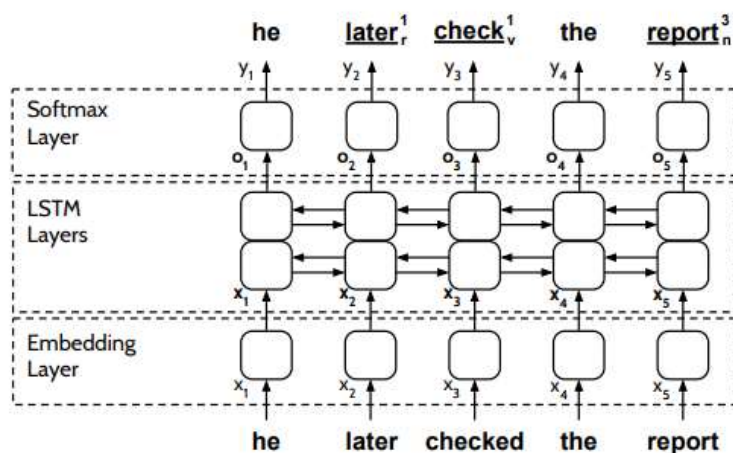


Automatic WSD Approaches

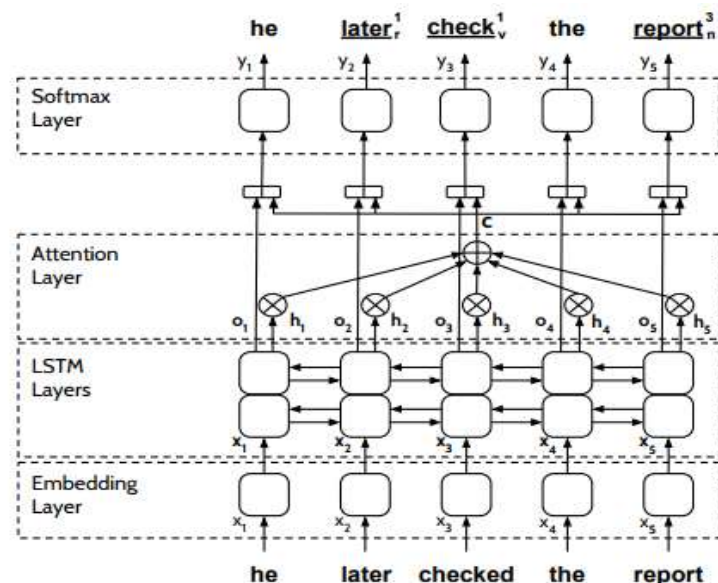
- **Knowledge Based Approach**
 - WSD using Selectional Preferences (or restrictions)
 - Overlap Based Approaches
- **Machine Learning Based Approach**
 - Supervised Approaches
 - Semi-supervised Algorithms
 - Unsupervised Algorithms
- **Hybrid Approach**
- **Neural Approach**

Neural Approach

- Raganato A et al. (2017)
 - Propose and study in depth a series of end-to-end neural architectures directly tailored to the task, from bidirectional Long Short-Term Memory to encoder-decoder models.



Bidirectional LSTM sequence labeling architecture for WSD (2 hidden layers).



Attentive bidirectional LSTM sequence labeling architecture for WSD (2 hidden layers).

Neural Approach

- Raganato A et al. (2017)

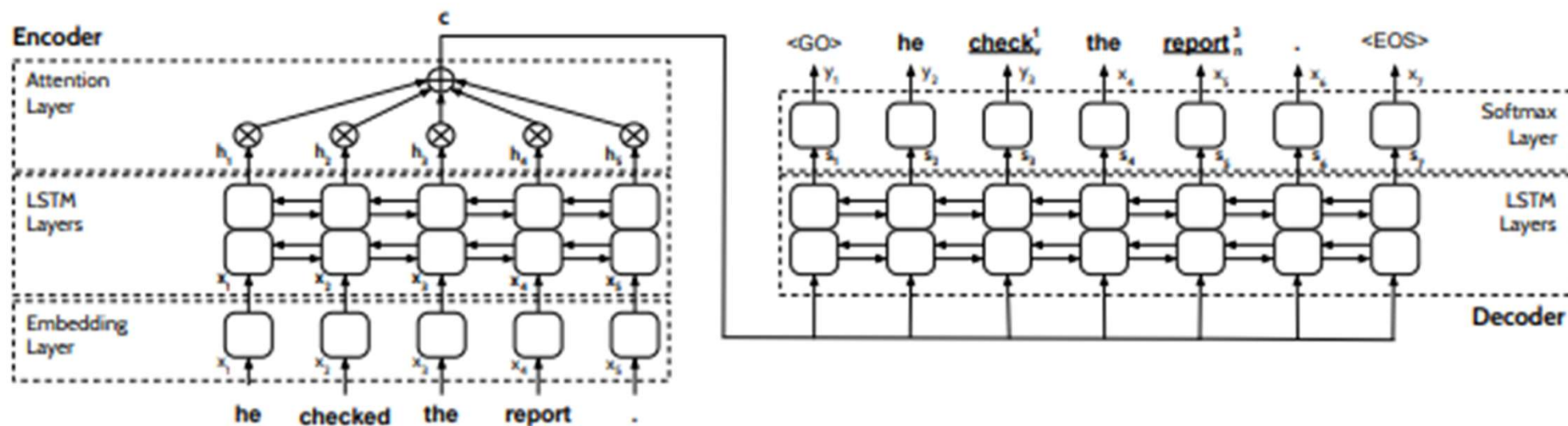
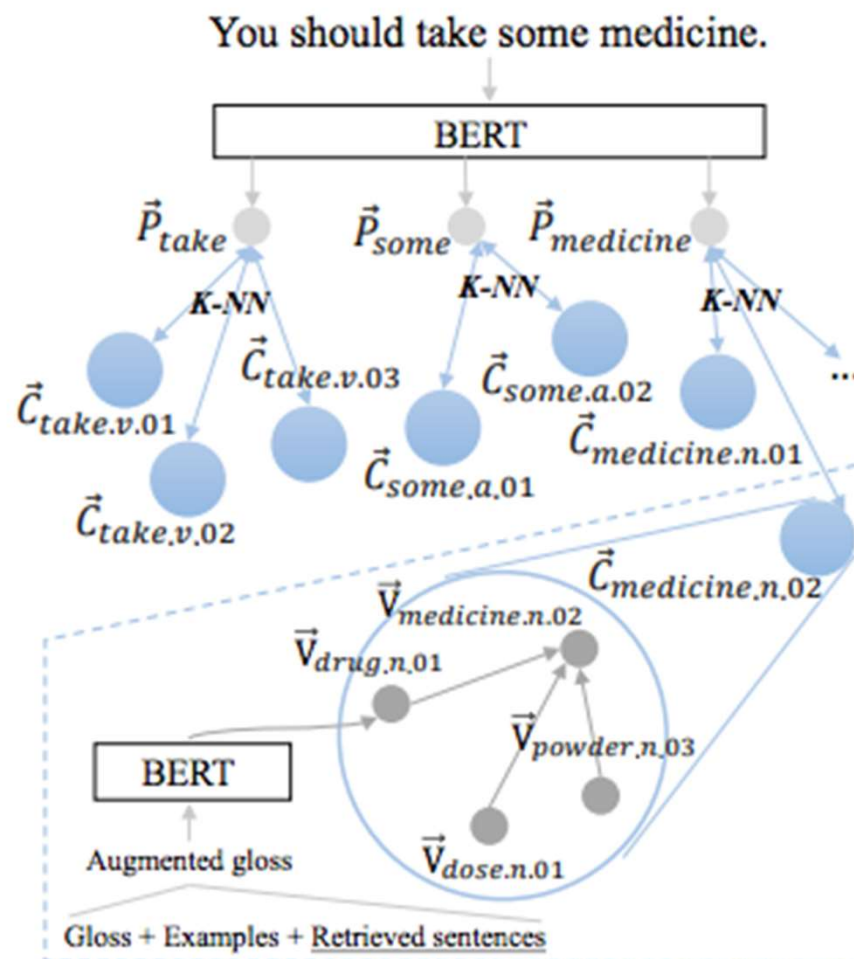


Figure 3: Encoder-decoder architecture for sequence-to-sequence WSD, with 2 bidirectional LSTM layers and an attention layer.

Neural Approach

- Contextual embeddings
 - [Wang et al. 2020] propose a Synset Relation-Enhanced Framework (SREF) that leverages sense relations for both sense embedding enhancement and a try-again mechanism that implements WSD again, after obtaining basic sense embeddings from augmented WordNet glosses.



Overcoming Knowledge Bottle-Neck



- Using Search Engines
 - Construct search queries using monosemic words and phrases from the gloss of a synset.
 - Feed these queries to a search engine.
 - From the retrieved documents extract the sentences which contain the search queries
- Using Equivalent Pseudo Words
 - Use monosemic words belonging to each sense of an ambiguous word.
 - Use the occurrences of these words in the corpus as training examples for the ambiguous word



Bounds

- Measure of how well the algorithm performs relative to the difficulty of the task.
- **Upper Bound**: Human performance.
- **Lower Bound or baseline**: Usually the assignment of all contexts to the most frequent sense.



The next lecture

- Lecture 8 Parsing I